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Impact of the Coronavirus Disease Pandemic on the hemorrhagic stroke: A Systematic Review and Meta-analysis

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Impact of the Coronavirus Disease Pandemic on the hemorrhagic stroke: A Systematic Review and Meta-analysis

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Abstract

Background and Purpose: COVID-19 pandemic, a global health crisis, disrupting the present medical environment. This systematic review and meta-analysis aimed to evaluate the impact of the coronavirus disease (COVID-19) pandemic on stroke, especially hemorrhagic stroke.

Methods: From the inception to January 29th 2021, the EMBASE, PubMed and Web of Science electronic databases were searched for all relevant studies. The data of interest were the number of stroke, hemorrhagic stroke, intracerebral hemorrhage and subarachnoid hemorrhage patients during the COVID-19 pandemic versus that during the pre-pandemic period.

Results: A total of 9 studies with 8553 patients were included. Overall, stroke admissions were lower in pandemic period vs. control period (3715 vs. 4614). The difference of hemorrhagic stroke is not significant, with 568 of 3715 patients (15.29%) in the pandemic group and 619 of 4614 patients (13.42%) in the control group (OR 1.11, 95% CI 0.98-1.26, I^2=12%, P=0.10). In subgroup analysis, intracerebral hemorrhage was present in 89 of 781 patients (11.40%) in the pandemic group and 74 of 903 patients (8.2%) in the control group, with statistically significant differences between two groups (OR 1.41, 95% CI 1.01-1.95, I^2=24%, P=0.04). As for subarachnoid hemorrhage, our results showed that the difference between two groups was not significant.

Conclusions: The number of stroke patients was lower in the pandemic period compared with the control period. There is a higher rate of hemorrhagic stroke in the pandemic period, but the difference was not significant. Subgroup analysis identified a significant increase in the

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occurrence of intracerebral hemorrhage in the pandemic period. Due to limited data, the impact of COVID-19 pandemic on subarachnoid hemorrhage is unclear.

Keywords: COVID-19, SARS-CoV-2, stroke, intracerebral hemorrhage, hemorrhagic stroke

Strengths and limitations of this study: 1) Because we collected results from different regions, the meaning of our study is wide and important. 2) The result of this study is more precise by meta-analysis, it could further explain disputes between different articles. 3) This study has the lack of information about stroke severity and prognoses, which might have to some extent underestimated the impact of pandemic for stroke.4) Total number of inpatients in the pandemic was not reported, so the prevalence of stroke is inconclusive. 5) Although our meta-analysis indicates a higher morbidity of ICH, the reliability needs to be verified by further researches because of the small sample size (163 patients).

Introduction

In December 2019, the 2019 novel coronavirus (COVID-19 or SARS-COV-2) was first reported in Wuhan, China[1]. On 30th January 2020, the World Health Organization (WHO) declared COVID-19 to become a Public Health Emergency of International Concern[2]. As of 19th January 2021, there have been over 93 million reported cases and over 2 million deaths globally, which has resulted in serious social and economic implications[3, 4].

During the pandemic, a lot of people rejected to visit the hospital due to the nosocomial infection, and some studies indicated that the incidence

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of acute conditions such as myocardial infarction and ischemic stroke was reduced[5, 6]. In addition, concerns have been raised about the impact of COVID-19 pandemic on hemorrhagic stroke (HS), mainly including intracerebral hemorrhage (ICH) and subarachnoid hemorrhage (SAH), which is controversial[7-9].

To further examine whether this pandemic influenced the admission of HS, we reviewed retrospective observational studies that compared rates of HS admissions between corresponding periods in 2020 and 2019.

Materials and methods

Research strategy and selection criteria

Our systematic review was conducted in accordance with the PRISMA guidelines. Two authors (YJY and YLN) independently identified studies published until January 29th 2021 through systematically searching EMBASE, PubMed and Web of Science. The following search key terms were used: cerebral hemorrhage, COVID-19, SARS-CoV-2, pandemic. We used Boolean operators "AND" or "OR" to combine the literature searches. No language restrictions were applied. Every eligible publication was selected based on articles' title and abstract by two authors (YJY and YLN). If there was any disagreement during study screening for relevance, we discussed with a senior author (PYD). This systematic review and meta-analysis was not registered in the International Prospective Register of Systematic Reviews database.

The inclusion criteria for this review were original articles and research letters. The data of interest were the number of ICH, SAH, HS and

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stroke patients during the COVID-19 pandemic versus the comparator period. Only studies involved with adults were eligible for inclusion. The exclusion criteria included conference abstracts, preprints, case reports, review articles, biochemical trials, no appropriate outcomes or trials without proper treatment groups and control groups.

Data extraction and quality assessment

The following data were extracted from each article by two independent researchers (YJY and YLN): author name, publication year, study type and design, source and region of study, patient demographics (age, gender), number of intracerebral hemorrhage patients, number of subarachnoid hemorrhage patients, number of hemorrhagic stroke patients and number of stroke patients. Any conflicts between the researchers were resolved by discussion or decision of a third senior researcher. If there were incomplete data, we attempted to contact the corresponding authors for complete data. We also calculated the necessary information which was not reported in articles.

The Newcastle-Ottawa scale was used to evaluate the risk of bias of the included retrospective observational studies. The score consisted of eight items, including adequate case definition, representativeness of cases, selection of controls, definition of controls, comparability, ascertainment of exposure, same method and non-response rate. Total quality score ranged between 0 and 9. Studies assessed with ≥5 points were regarded as low risk of bias, compared to the studies scoring <5 points that were regarded as high risk of bias.

Data synthesis and analysis

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Meta-analysis of proportion was used to determine the number of ICH, SAH, HS and stroke patients during the pandemic compared to that during the historical pre-pandemic control period. The present study was performed by using Review Manager (Version 5.3.0) software. Heterogeneity was calculated by using Q test and I^2, in which $P<.1$, $I^2>50\%$ meant significant heterogeneity. A fixed-effect model was used when statistically indicated heterogeneity was found ($P>.1$, $I^2<50\%$). Sensitivity analysis was used to exclude the origins of heterogeneity. Because of the limited number of studies (<10), assessment for publication bias and meta-regression analysis was not performed.

Results

Study selection

By using the key phrases mentioned above, the literature search yielded 670 articles from the following databases: 114 from PubMed, 402 from EMBASE, 154 from Web of Science. 336 articles were retained after removal of duplicates. A total of 320 records were excluded after screening the titles and abstracts, leaving 16 eligible studies. After the full texts were assessed for eligibility, we excluded 7 articles for the following reasons: 1) Child (n=1), 2) inappropriate outcome (n=5), 3) ischemic stroke (n=1). Finally, 9 studies were eligible and included in our systematic review[8-16]. A PRISMA diagram detailing the study selection process is shown in Figure 1.

Characteristics of included studies

The main characteristics of included studies, including author, the design of study, country or region, age, male to female ratio, pandemic period,

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control period and the number of each disease are summarized in Table 1.

A total sample of 8553 subjects from 9 studies was included in the systematic review. The studies were conducted in many countries worldwide including Italy[10, 13], United Arab Emirates[11], Finland[12], USA[9], Germany[14], Ghana[15], Italy[8] and Egypt[16]. All of the studies were retrospective-observational and reported on the number of ICH, SAH, HS and stroke patients during the COVID-19 pandemic compared to the control period.

Risk of bias within studies

The Newcastle-Ottawa score results for each study are shown in table 2. All of these studies were assessed as having a low risk of bias.

Synthesis of results

Hemorrhagic stroke

The number of HS was reported in 8 articles[8-11, 13-16]. The pandemic group tended to have a higher rate of HS, with 568 of 3715 patients (15.29%) in the pandemic group and 619 of 4614 patients (13.42%) in the control group. However, the difference was not significant (OR 1.11, 95% CI 0.98-1.26, I^2=12%, P=0.10) (Figure 2A).

Subgroup analysis

On the basis of the type of HS, a subgroup analysis was performed to discuss some outcomes. In ICH, the number was reported in 5 articles[9, 11,

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[13, 14, 16]. It was present in 89 of 781 patients (11.40%) in the pandemic group and 74 of 903 patients (8.2%) in the control group, with statistically significant differences between two groups (OR 1.41, 95% CI 1.01-1.95, I^2=24%, P=0.04) (Figure 2B). In SAH, Only 3 articles showed the number of SAH patients[9, 11, 16]. 14 of 350 patients (4.0%) in the pandemic group and 16 of 379 patients (4.22%) in the control group suffered from SAH. The difference was not significant (OR 0.66, 95% CI 0.31-1.41, I^2=0%, P=0.29) (Figure 2C).

Sensitivity analysis

Sensitivity analysis showed that no individual study had a significant impact on the outcomes.

Discussion

In this review, the results show that the number of stroke patients was reduced in COVID-19 pandemic. Although the number of HS was less during the pandemic period than during the pre-pandemic period, the proportion of HS in stroke was no significant change. Our subgroup analysis further indicates that ICH patients increased from 74 in 2019 to 89 in 2020, with a corresponding rate raised. And there is no significant difference in SAH.

There are many previous studies found that the COVID-19 outbreak observed the decline in stroke admissions in Italy, Spain, Hong Kong, France, China and the USA[8, 17-20]. But in Ghana, more than previous number of stroke patients has been reported in the outbreak of COVID-19 pandemic[15]. The difference may be related to governmental policies, medical level and the levels of transmission. Our results support the

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decreased trend in stroke during the pandemic period though analyzing 8329 subjects from 8 studies. We find three possible reasons to explain this phenomenon. Firstly, patients with stroke and mild symptoms preferred to stay at home and rejected to the emergency department for treatment in pandemic period. Secondly, increased social isolation decreased the discovery rate of stroke patients, because their friends or family members couldn't recognize who was suffering from a stroke. Thirdly, under city lockdown, limited transportation made some patients hardly to acquire ambulance and drives, which delayed the treatment for stroke. So we appeal for that local government should carry out focused education and intervention for high-risk group and provide a green channel for the treatment of strokes. On the other hand, people should strengthen connection with relatives and friends by phone or online video.

Although we document a numerically lower number of HS, the difference is not statistically significant. Similarly, Balestrino et al reported that the number of HS did not change at all during the pandemic[10]. But in our subgroup analysis, we notice that ICH had a significant increase in the occurrence during the epidemics. In a previous review and meta-analysis of risk of factors for spontaneous ICH, we identified hypertension and alcohol intake as important risk factors[21]. We further find some studies reported that pandemic has led to an increase in alcohol consumption and alcohol abuse which may be attributed to the stress and isolation experienced with the current pandemic[22-32]. Therefore, we think that the significant increase of ICH may be related to alcohol abuse and stress during the pandemic. But the authenticity needs further validation by special studies. In addition, the data variability is small, because ICH patients kept seeking hospital care in emergency medical services in the

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2020 as they had done in 2019 (e.g. The emergency way heavily avoided that some mild patients choose to keep away from hospitals). In summary, in this particular period, the quarantine and social isolation just like some degree of grouping which exclude other potentially confounding risk factors. It makes us have gained a deeper appreciation of obvious risk factors for ICH. In view of the above-mentioned facts, we wish to make the following proposals: 1) People should pay more attention to modifiable risk factors such as alcohol intake and hypertension to reduce the risk of ICH in the COVID-19 pandemic period. 2) Various health organizations and clinicians should develop recommendations for people on how to use healthy lifestyle to cope with physical distancing and social isolation. 3) Governments should explore effective prevention and intervention measures to prevent crises. As for SAH, we think there is too little data to draw any important conclusions. It needs more authoritative studies to explore the impact of COVID-19 pandemic on SAH.

Limitations

Limitation to this study can be found in the lack of information about stroke severity and prognoses, which might have to some extent underestimated the impact of pandemic for stroke. Moreover, total number of inpatients in the pandemic was not reported, so the prevalence of stroke is inconclusive. Although our meta-analysis indicates a higher morbidity of ICH, the reliability needs to be verified by further researches because of the small sample size (163 patients). Despite this, this study still has notable clinical and public health implications. For example, our data may encourage people to conduct further research for testifying the potential pathogenesis of ICH. On the other hand, our results could

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attract more attention to impacts of COVID-19 pandemic, so people can prefect isolation approaches and prevention measures.

Conclusions

The study identified moderate reductions of stroke admissions during the COVID-19 pandemic. Alcohol abuse and stress may be important causes of increasing ICH events at this special time. All governments, healthcare institutions, clinicians and academics should aim at preventing potential crisis in the COVID pandemic, exploring more effective prevention and intervention measures.

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Author Contributions: YJY conceived and designed the study and drafted the manuscript. YJY and YLN acquired the data and performed data extraction. YJY interpreted the data, and performed extensive research on the topic. YJY performed the statistical analysis and made all tables and figures. All authors contributed to the accomplishment of the manuscript.

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Conflicts of interest: The authors declare that they have no competing interests.

References

1. Huang C, Wang Y, Li X, Ren L, Zhao J, Hu Y, et al. Clinical features of patients infected with 2019 novel coronavirus in wuhan, china. *The Lancet*. 2020;395:497-506
2. Sohrabi C, Alsafi Z, O'Neill N, Khan M, Kerwan A, Al-Jabir A, et al. World health organization declares global emergency: A review of the 2019 novel coronavirus (covid-19). *International journal of surgery (London, England)*. 2020;76:71-76

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3. WHO. Coronavirus disease (covid-19) weekly epidemiological update - 19 january 2021. 2021
4. Lipsitch M, Swerdlow DL, Finelli L. Defining the epidemiology of covid-19 - studies needed. *The New England journal of medicine*. 2020;382:1194-1196
5. Solomon MD, McNulty EJ, Rana JS, Leong TK, Lee C, Sung SH, et al. The covid-19 pandemic and the incidence of acute myocardial infarction. *The New England journal of medicine*. 2020;383:691-693
6. July J, Pranata R. Impact of the coronavirus disease pandemic on the number of strokes and mechanical thrombectomies: A systematic review and meta-analysis. *Journal of stroke and cerebrovascular diseases : the official journal of National Stroke Association*. 2020;29:105185
7. Wang X, Dong Y, Qi X, Huang C, Hou L. Cholesterol levels and risk of hemorrhagic stroke: A systematic review and meta-analysis. *Stroke*. 2013;44:1833-1839
8. Sacco S, Ricci S, Ornello R, Eusebi P, Petraglia L, Toni D. Reduced admissions for cerebrovascular events during covid-19 outbreak in italy. *Stroke*. 2020;51:3746-3750
9. Rameez F, McCarthy P, Cheng Y, Packard LM, Davis AT, Wees N, et al. Impact of a stay-at-home order on stroke admission, subtype, and metrics during the covid-19 pandemic. *Cerebrovascular diseases extra*. 2020;10:159-165
10. Balestrino M, Coccia A, Boffa AS, Furgani A, Bermano F, Finocchi C, et al. Request of hospital care dropped for tia but remained stable for stroke during covid-19 pandemic at a large italian university hospital. *Internal and emergency medicine*. 2020:1-5
11. John S, Hussain SI, Piechowski-Jozwiak B, Dibu J, Kesav P, Bayrlee A, et al. Clinical characteristics and admission patterns of stroke patients during the covid 19 pandemic: A single center retrospective, observational study from the abu dhabi, united arab emirates. *Clinical neurology and neurosurgery*. 2020;199:106227
12. Luostarinen T, Virta J, Satopää J, Bäcklund M, Kivisaari R, Korja M, et al. Intensive care of traumatic brain injury and aneurysmal subarachnoid hemorrhage in helsinki during the covid-19 pandemic. *Acta neurochirurgica*. 2020;162:2715-2724
13. Zini A, Romoli M, Gentile M, Migliaccio L, Picoco C, Dell'Arciprete O, et al. The stroke mothership model survived during covid-19 era: An observational single-center study in emilia-romagna, italy. *Neurological sciences : official journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology*. 2020;41:3395-3399
14. Schwarz V, Mahfoud F, Lauder L, Reith W, Behnke S, Smola S, et al. Decline of emergency admissions for cardiovascular and cerebrovascular events after the outbreak of covid-19. *Clinical research in cardiology : official journal of the German Cardiac Society*. 2020;109:1500-1506
15. Sarfo FS, Mensah NO, Opoku FA, Adusei-Mensah N, Ampofo M, Ovbiagele B. Covid-19 and stroke: Experience in a ghanaian healthcare system. *Journal of the neurological sciences*. 2020;416:117044
16. Roushdy TM, Nahas NME, Aref HM, Georgy SS, Zaki AS, Bedros RY, et al. Stroke in the time of coronavirus disease 2019: Experience of two university stroke centers in egypt. *Journal of stroke*. 2020;22:275-277
17. Yaghi S, Ishida K, Torres J, Mac Grory B, Raz E, Humbert K, et al. Sars-cov-2 and stroke in a new york healthcare system. *Stroke*. 2020;51:2002-2011
18. Zhao J, Li H, Kung D, Fisher M, Shen Y, Liu R. Impact of the covid-19 epidemic on stroke care and potential solutions. *Stroke*. 2020;51:1996-2001
19. Rudilosso S, Laredo C, Vera V, Vargas M, Renú A, Llull L, et al. Acute stroke care is at risk in the era of covid-19: Experience at a comprehensive stroke center in

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barcelona. *Stroke*. 2020;51:1991-1995

20. Teo KC, Leung WCY, Wong YK, Liu RKC, Chan AHY, Choi OMY, et al. Delays in stroke onset to hospital arrival time during covid-19. *Stroke*. 2020;51:2228-2231
21. Ariesen MJ, Claus SP, Rinkel GJ, Algra A. Risk factors for intracerebral hemorrhage in the general population: A systematic review. *Stroke*. 2003;34:2060-2065
22. Ćosić K, Popović S, Šarlija M, Kesedžić I. Impact of human disasters and covid-19 pandemic on mental health: Potential of digital psychiatry. *Psychiatria Danubina*. 2020;32:25-31
23. Sun Y, Li Y, Bao Y, Meng S, Sun Y, Schumann G, et al. Brief report: Increased addictive internet and substance use behavior during the covid-19 pandemic in china. *The American journal on addictions*. 2020;29:268-270
24. Colbert S, Wilkinson C, Thornton L, Richmond R. Covid-19 and alcohol in australia: Industry changes and public health impacts. *Drug and alcohol review*. 2020;39:435-440
25. Kmiec J. President's message: The impact of the covid-19 public health emergency on the practice of addiction medicine. *Journal of addictive diseases*. 2020;38:235-236
26. Andrade C. Covid-19 and lockdown: Delayed effects on health. *Indian journal of psychiatry*. 2020;62:247-249
27. Rehm J, Kilian C, Ferreira-Borges C, Jernigan D, Monteiro M, Parry CDH, et al. Alcohol use in times of the covid 19: Implications for monitoring and policy. *Drug and alcohol review*. 2020;39:301-304
28. Carrico AW, Horvath KJ, Grov C, Moskowitz JT, Pahwa S, Pallikkuth S, et al. Double jeopardy: Methamphetamine use and hiv as risk factors for covid-19. *AIDS and behavior*. 2020;24:3020-3023
29. Clay JM, Parker MO. Alcohol use and misuse during the covid-19 pandemic: A potential public health crisis? *The Lancet. Public health*. 2020;5:e259
30. Shigemura J, Ursano RJ, Morganstein JC, Kurosawa M, Benedek DM. Public responses to the novel 2019 coronavirus (2019-ncov) in japan: Mental health consequences and target populations. *Psychiatry and clinical neurosciences*. 2020;74:281-282
31. Wang Y, Di Y, Ye J, Wei W. Study on the public psychological states and its related factors during the outbreak of coronavirus disease 2019 (covid-19) in some regions of china. *Psychology, health & medicine*. 2021;26:13-22
32. Ramalho R. Alcohol consumption and alcohol-related problems during the covid-19 pandemic: A narrative review. *Australasian psychiatry : bulletin of Royal Australian and New Zealand College of Psychiatrists*. 2020;28:524-526

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Figure legends

Figure 1. Flow chart of study selection process

Figure 2. (A) Forest plot showing the number of HS during the pandemic compared with the control period. (B) Forest plot showing the number of ICH during the pandemic compared with the control period. (C) Forest plot showing the number of SAH during the pandemic compared with the control period.

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Table 1: Summary of characteristics of included studies.

Author	Study Design	Region	Age(years)	Male(%)	Pandemic Period	Control Period	Total number of Intracerebral hemorrhage patients (2020 vs Control)	Total number of subarachnoid hemorrhage patients (2020 vs Control)	Total number of hemorrhagic stroke patients (2020 vs Control)	Total number of stroke patients (2020 vs Control)
Balestrino et al, 2020	Single Center; Retrospective Observational	Italy	N/A*	N/A*	8st March 2020 to 2st May 2020	8st March 2019 to 2st May 2019	N/A*	N/A*	20 vs 17	124 vs 165
John et al, 2020	Single Center; Retrospective Observational	United Arab Emirates	48.9(±14.7) vs 49.3(±14.1)	66.7% vs 80.8%	1st March 2020 to10st May 2020	1st March 2019 to10st May 2019	24 vs 12	13 vs 13	42 vs 26	210 vs 148
Luostarinen, 2020	Single Center; Retrospective Observational	Finland	50(62-72) vs 56(44-71)	62% vs 58%	1st January 2020 to 31st May 2020	1st January 2020 to 31st May 2019	67 vs 56	52 vs 49	N/A*	N/A*
Rameez et al, 2020	Single Center; Retrospective Observational	USA	65.6 vs 72.4	61.7% vs 51.3%	24st March 2020 to 23st April 2020	24st March 2019 to 23st April 2019	9 vs 12	0 vs 0	9 vs 12	47 vs 80
Zini et al, 2020	Single Center; Retrospective Observational	Italy	71.7(±18) vs 75.2(±13.8）	53.8% vs 50.7%	1st March 2020 to 30st April 2020	1st March 2019 to 30st April 2019	17 vs 21	N/A*	17 vs 21	145 vs 138
Schwarz et al, 2020	Single Center; Retrospective Observational	Germany	N/A*	N/A*	Calendar weeks 1 to 16 in 2020	Calendar weeks 1 to 16 in 2019	32 vs 24	N/A*	32 vs 24	286 vs 386

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Sarfo et al, 2020	Single Center; Retrospective Observational	Ghana	60.6(±14.8) vs 59.7(±15.4）	57.1% vs 53.4%	19st January 2020 to 20st June 2020	19st January 2019 to 20st June 2019	N/A*	N/A*	118 vs 111	431 vs 401
Sacco et al, 2020	Multicenter; Prospectiveand Retrospective	Italy	73.5(±12.9) vs 73.6(±13.3）	53% vs 53.4%	1st March 2020 to 31st March 2020	1st March 2019 to 31st March 2019	N/A*	N/A*	322 vs 400	2379 vs 3145
Roushdy et al, 2020	Single Center; Retrospective Observational	Egypt	63.2(±12.7) vs 63(±13.8）	61.3% vs 57%	15st February 2020 to 5st April 2020	15st February 2019 to 5st April 2019	7 vs 5	1 vs 3	8 vs 8	93 vs 151

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Table 2: Newcastle-Ottawa quality assessment scale and total score of each study.

Included Studies	Newcastle-Ottawa quality assessment scale								Score
	1	2	3	4	5	6	7	8	Total of number of stars
Balestrino et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
John et al, 2020	A*	A*	B	A*	A*	A*	A*	A*	7
Luostarinen, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Rameez et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Zini et al, 2020	A*	A*	B	A*	A*	A*	A*	A*	7
Schwarz et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Sarfo et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Sacco et al, 2020	A*	A*	A*	A*	A**	A*	A*	A*	9
Roushdy et al, 2020	B	B	B	A*	A*	A*	A*	A*	5

Note: A*, one star; A, two stars; B, no star; 1, Adequate Case Definition; 2, Representativeness of Cases; 3, Selection of Controls; 4, Definition of Controls; 5, Comparability; 6, Ascertainment of Exposure; 7, Same Method; 8, Non-Response Rate.**

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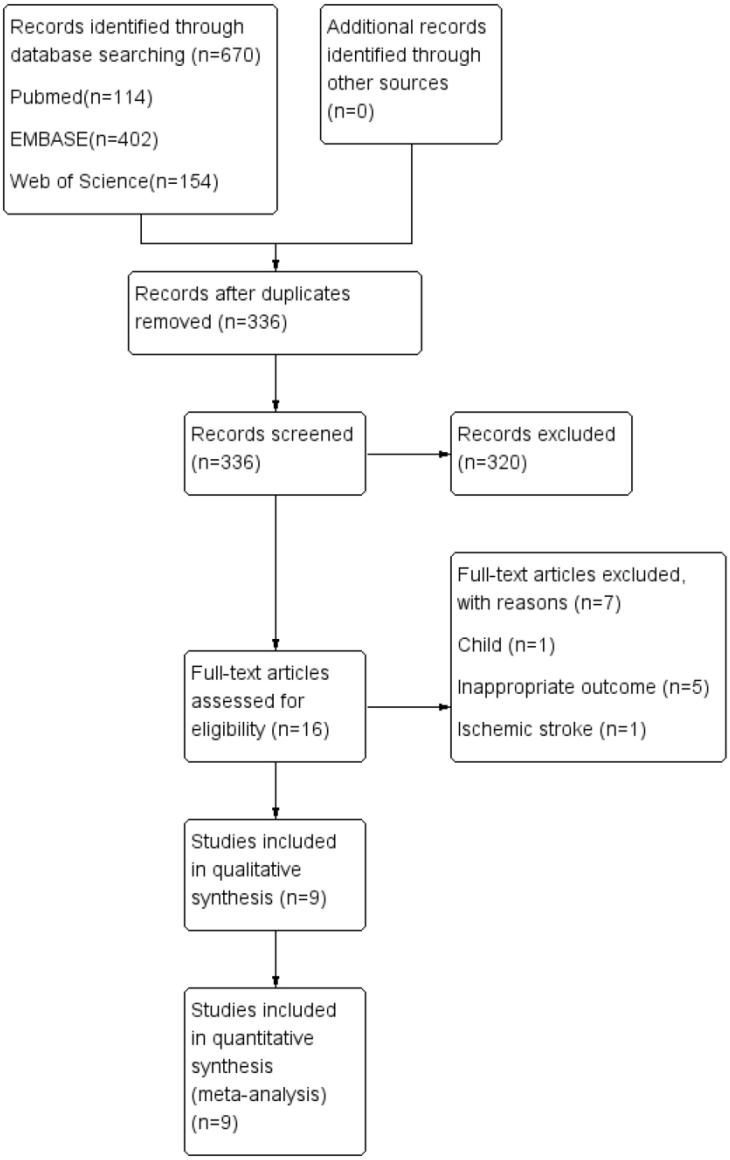


Figure 1. Flow chart of study selection process

170x264mm (300 x 300 DPI)

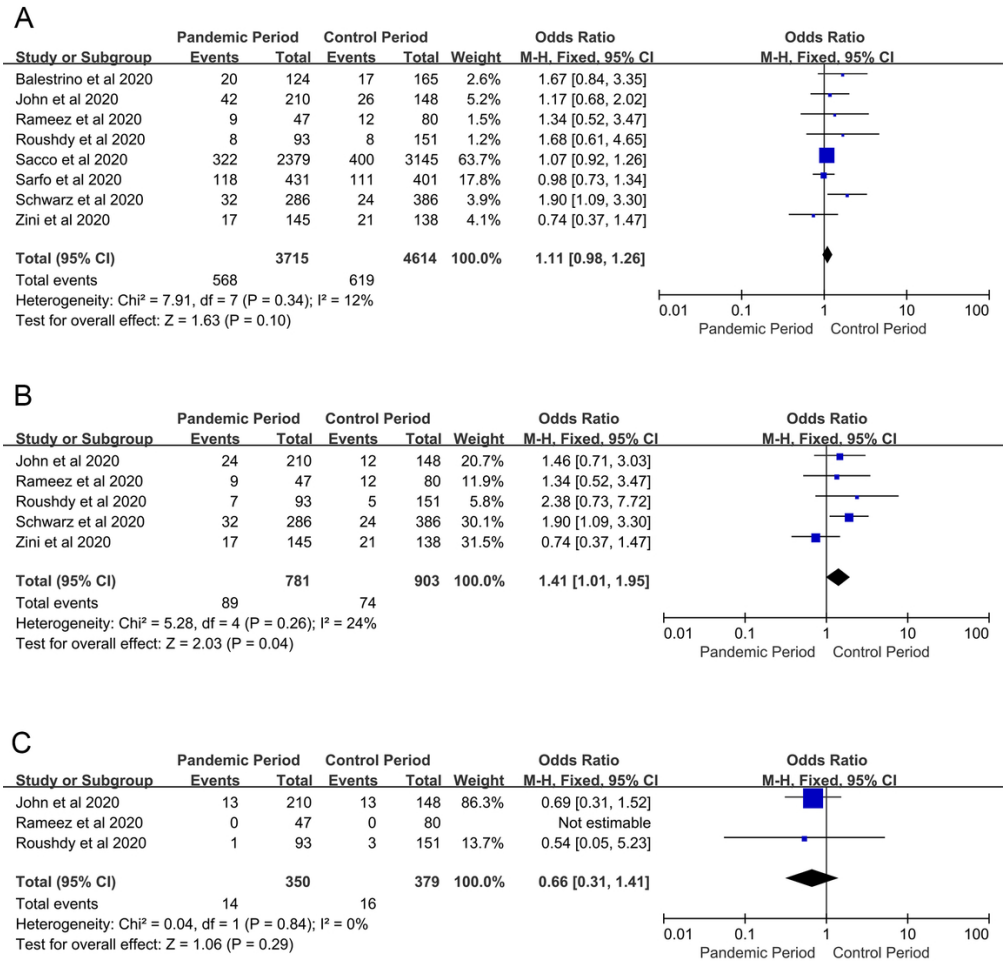


Figure 2. (A) Forest plot showing the number of HS during the pandemic compared with the control period. (B) Forest plot showing the number of ICH during the pandemic compared with the control period. (C) Forest plot showing the number of SAH during the pandemic compared with the control period.

48x48mm (600 x 600 DPI)



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	3-4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	3-4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5-6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6-7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	-
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7-9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9-10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	10

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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The impact of COVID-19 Pandemic on Hemorrhagic Stroke Admissions: A Systematic Review and Meta-analysis

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The impact of COVID-19 Pandemic on Hemorrhagic Stroke Admissions: A Systematic Review and Meta-analysis

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Abstract

Background and Purpose: COVID-19 pandemic, a global health crisis, disrupting the present medical environment. This systematic review and meta-analysis aimed to evaluate the impact of the coronavirus disease (COVID-19) pandemic on hospitalizations of stroke, especially hemorrhagic stroke.

Methods: The EMBASE, PubMed, Web of Science, Elsevier, Medline, Cochrane Library and Google scholar electronic databases were searched for all relevant studies. Two researchers independently screened the studies, extracted data and assessed the quality of the included studies. OR, total events, odds ratio and 95% CI were considered as the effect size. A fixed effects model was used to pool the study-specific estimate. The present study was performed by using Review Manager (Version 5.3.0) software. We assessed the risk of bias using the Newcastle-Ottawa Scale.

Results: A total of 17 studies with 7608 cases were included. Overall, the number of stroke admissions is lower in the pandemic period vs. the control period (6252 vs. 8193). The difference of hemorrhagic stroke is significant, with 1233 of 6252 cases in the pandemic group and 1621 of 8193 cases in the control group. Intracerebral hemorrhage is present in 461 of 1948 cases in the pandemic group and 618 of 2734 cases in the control group. As for subarachnoid hemorrhage, the difference between the two groups is significant, with 70 of 985 cases in the pandemic group and 202 of 1493 cases in the control group.

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Conclusions: The number of stroke admissions is lower in the pandemic period compared with the control period. There is a higher rate of hemorrhagic stroke in the pandemic period. Subgroup analysis identifies a significant increase in the occurrence of intracerebral hemorrhage in the pandemic period. Due to limited data and the impact of a single article, the impact of COVID-19 pandemic on subarachnoid hemorrhage is unclear.

Keywords: COVID-19, SARS-CoV-2, stroke, intracerebral hemorrhage, hemorrhagic stroke

Strengths and limitations of this study: 1) Because we collected results from different regions, the meaning of our study is wide and important. 2) The result of this study is credible by meta-analysis, which could further explain disputes between different articles. 3) This study lacks information about stroke severity and prognoses, which might have to some extent underestimated impacts of the pandemic for stroke. 4) The total number of inpatients in the pandemic was not reported, so the prevalence of stroke is inconclusive. 5) Although our meta-analysis indicates higher morbidity of ICH, the reliability needs to be verified by further researches because of the small sample size.

Introduction

In December 2019, the 2019 novel coronavirus (COVID-19 or SARS-COV-2) was first reported in Wuhan, China[1]. On 30th January 2020, the World Health Organization (WHO) declared COVID-19 to become a Public Health Emergency of International Concern[2]. As of 16th August 2021, there have been over 21.2 million confirmed cases and 761000 deaths globally, resulting in serious social and economic implications[3, 4].

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Stroke is the second leading cause of death and functional inability worldwide, and one of the main contributors of disability adjusted life years among neurological disorders[5, 6]. The burden of stroke is projected to increase via absolute numbers of incident strokes and deaths[7]. The main contributor to stroke burden is ischemic stroke, followed by intracerebral hemorrhage (ICH) and subarachnoid hemorrhage (SAH)[7, 8]. To prevent and treat stroke, most countries try to establish major and emerging strategies, including activities that reinforce the local healthcare system through the establishment of a social, economic, environmental and cultural combination[9, 10].

Unfortunately, the outbreak of COVID-19 affecting acute stroke services, both direct and indirect. During the pandemic, a lot of people refuse to visit the hospital due to nosocomial infection, and some studies indicate that the incidence of acute conditions such as myocardial infarction and ischemic stroke is reduced[11, 12]. The WSO survey showed that the number of acute stroke admissions was reduced in multiple countries, including Chile, Colombia, Iran, Green, UK, Belgium and Italy[13]. In addition, concerns have been raised about the impact of COVID-19 pandemic on hemorrhagic stroke (HS), mainly including ICH and SAH[14-16]. For example, Balestrino et al., John et al. and Schwarz et al. reported that the admission of hemorrhagic stroke was increased in the pandemic period compared to the control period. However, reports of Rameez et al., Zini et al. and Sacco et al. showed that the admission of hemorrhagic stroke was reduced in the pandemic period. It makes the impact remain controversial. Moreover, most of these reports are limited to regional or country-specific analysis, and thus, there is no systematic review and meta-analysis on the impact of the COVID-19 pandemic on the admission of HS. To further examine whether this pandemic

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influenced the admission of HS, we reviewed recent studies that compared rates of HS admissions between corresponding periods in pandemic and pre-pandemic.

Materials and methods

Research strategy and selection criteria

Our systematic review was conducted in accordance with the PRISMA guidelines. Two authors (YJY and YLN) independently identified studies published until August 22th 2021 through systematically searching EMBASE, PubMed, Web of Science, Elsevier, Medline, Cochrane Library and Google scholar. The following search key terms were used: stroke, subarachnoid hemorrhage, hemorrhagic stroke, cerebral hemorrhage, COVID-19, SARS-CoV-2, pandemic. We used Boolean operators "AND" or "OR" to combine the literature searches (supplementary appendix 1). No language restrictions were applied. Every eligible publication was selected based on the articles' title and abstract by two authors (YJY and YLN). If there was any disagreement during study screening for relevance, we discussed it with a senior author (PYD). This systematic review and meta-analysis was not registered in the International Prospective Register of Systematic Reviews database.

The inclusion criteria for this review were original articles, retrospective observational studies, prospective and retrospective studies, and research letters. The data of interest were the admission of ICH, SAH, HS and stroke during the COVID-19 pandemic versus the comparator period. Only studies involved with adults were eligible for inclusion. The exclusion criteria included protocols, retracted articles, conference

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abstracts, preprints, case reports, review articles, biochemical trials, no appropriate outcomes or trials without proper treatment groups and control groups.

Data extraction and quality assessment

The following data were extracted from each article by two independent researchers (YJY and YLN): author name, publication year, study type and design, source and region of study, patient demographics (age, gender), number of intracerebral hemorrhage admissions, number of subarachnoid hemorrhage admissions, number of hemorrhagic stroke admissions and number of stroke admissions. Any conflicts between the researchers were resolved by discussion or decision of a third senior researcher. If there were incomplete data, we attempted to contact the corresponding authors for complete data. We also calculated the necessary information which was not reported in articles.

The Newcastle-Ottawa scale was used to evaluate the risk of bias of the included retrospective observational studies. The score consisted of eight items, including adequate case definition, representativeness of cases, selection of controls, definition of controls, comparability, ascertainment of exposure, same method and non-response rate. The total quality score ranged between 0 and 9. Studies assessed with ≥5 points were regarded as low risk of bias, compared to the studies scoring <5 points that were regarded as high risk of bias.

Data synthesis and analysis

Meta-analysis of proportion was used to determine the admission of ICH, SAH, HS and stroke during the pandemic compared to that during the

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historical pre-pandemic control period. The present study was performed by using Review Manager (Version 5.3.0) software. Heterogeneity was calculated by using Q test and I^2, in which $P<.1$, $I^2>50\%$ meant significant heterogeneity. A fixed-effect model was used when statistically indicated heterogeneity was not found ($P>.1$, $I^2<50\%$). Sensitivity analysis was used to assess the impact of individual articles. If there were a limited number of studies (<10), an assessment for publication bias was not performed. Conversely, publication bias was analyzed and represented by a funnel plot that allowed evaluation of publication bias by presenting the study's log OR as a function of its standard error.

Patient and public involvement

This is a meta-analysis based on study-level data and no individual-level data were involved in the study or in defining the research question or outcome measures.

Results

Study selection

By using the key phrases mentioned above, the literature search yielded 7608 articles from the following databases: 2319 from PubMed, 2079 from EMBASE, 1068 from Web of Science, 463 from Elsevier, 1572 from Medline, 87 from Cochrane Library, 10 from Google scholar, 10 from other sources. 1775 articles were retained after removal of duplicates. A total of 1708 records were excluded after screening the titles and abstracts, leaving 67 eligible studies. After the full texts were assessed for eligibility, we excluded 50 articles for the following reasons: 1)

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molecular biology studies (n=5), 2) protocol (n=1), 3) retracted article (n=1), 4) without relevant data (n=43). Finally, 17 studies were eligible and included in our systematic review[15-23]. A PRISMA diagram detailing the study selection process is shown in Figure 1.

Characteristics of included studies

The main characteristics of included studies, including author, the design of study, country or region, age, male to female ratio, pandemic period, control period and the admission of each disease are summarized in Table 1.

A total sample of 14445 subjects from 17 studies was included in the systematic review. The studies were conducted in many countries worldwide including Italy[15, 17, 20], United Arab Emirates[18], USA[16], Germany[21, 24, 25], Ghana[22], Turkey[26], Switzerland[27], Austria[28], Iran[29], Bangladesh[30], Spain[31, 32] and Egypt[23]. These studies included retrospective-observational studies and prospective and retrospective studies. All studies were reported on the number of ICH, SAH, HS and stroke admissions during the COVID-19 pandemic compared to the control period.

Risk of bias within studies

The Newcastle-Ottawa score results for each study were shown in Table 2. All of these studies were assessed as having a low risk of bias.

Synthesis of results

Hemorrhagic stroke

The admission of HS was reported in 17 articles[15-18, 20-28, 30-32]. In HS, the pandemic group has 1233 of 6252 cases and the control group has 1621

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of 8193 cases. The proportion of HS admissions is higher at stroke admissions in the pandemic period than the proportion of HS admissions in the control period. The difference is significant (OR 1.10, 95% CI 1.01-1.21, I^2=10%, P=0.03) (Figure 2A). Figure 3A shows a funnel plot for the visual inspection of publication bias. The plot shows there is no significant publication bias.

Subgroup analysis

To explore the reason for the increased proportion of HS admissions in the pandemic, we conducted a subgroup analysis based on disease categories. In ICH, the admission was reported in 11 articles[16, 18, 20, 21, 23-27, 29, 30]. It was present in 461 of 1948 cases in the pandemic group and 618 of 2734 cases in the control group, with statistically significant differences between two groups (OR 1.40, 95% CI 1.19-1.64, I^2=0%, P<0.0001) (Figure 2B). The funnel plot shows there is no significant publication bias (Figure 3B). In SAH, only 5 articles showed the admission of SAH[16, 18, 23]. 70 of 985 cases in the pandemic group and 202 of 1493 cases in the control group suffered from SAH. The difference is significant (OR 0.52, 95% CI 0.39-0.70, I^2=0%, P<0.0001) (Figure 2C).

Sensitivity analysis

Sensitivity analysis shows that the study of Hasan et al. has a significant impact on the outcomes of SAH. As for the outcomes of HS and ICH，there is no individual study that has a significant impact on the outcomes.

Discussion

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In this review, the result shows that the number of stroke admissions is reduced in the COVID-19 pandemic. Although the admissions of HS are less during the pandemic period than during the pre-pandemic period, the proportion of HS hospitalization in stroke hospitalization is significantly increased. Our results further indicate that the number of ICH admissions reduces from 618 in the pandemic group to 461 in the control group, with a corresponding rate raised. As for SAH, the number of SAH admissions and the proportion of SAH admissions are less during the pandemic period than during the control period, but sensitivity analysis indicates that exclusion of an article (Hasan et al.) influences the pooled estimates.

The WSO survey indicated that the COVID-19 pandemic had a momentous impact on stroke care with delayed presentation and reduced hospital admissions. Our results further support the decreased trend in stroke admissions during the pandemic period by analyzing 14445 cases from 17 studies. But our results on overall strokes might be not comprehensive, because we only considered studies that reported data on HS separately. Although different search strategies are used to weaken the study selection bias, we think that the bias is an objective existence. At all events, the phenomenon of reduced stroke admissions is certain. We find three possible reasons to explain this phenomenon. Firstly, patients with stroke and mild symptoms preferred to stay at home and rejected the emergency department for treatment in the pandemic period. Secondly, increased social isolation decreased the discovery rate of stroke patients, because their friends or family members could not recognize who was suffering from a stroke. Thirdly, under city lockdown, limited transportation made some patients hardly acquire ambulances and drives, which delayed the

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treatment for stroke. So we appeal that local government should carry out focused education and intervention for high-risk groups, and provide a green channel for the treatment of strokes. On the other hand, people should strengthen their connection with relatives and friends by phone or online video.

Additionally, we document a numerically higher proportion of HS admissions in stroke admissions and notice that ICH has a significant increase in occurrence during the epidemics. In a previous review and meta-analysis of the risk factors for spontaneous ICH, we identified hypertension and alcohol intake as important risk factors[33]. We further found some studies reported that pandemic led to an increase in alcohol consumption and alcohol abuse which may be attributed to the stress and isolation experienced with the current pandemic[34-44]. Therefore, we think that the significant increase of ICH is related to alcohol abuse and stress during the pandemic. But the authenticity needs further validation by special studies. In addition, the data variability is small, because ICH patients kept seeking hospital care in emergency medical services in the pandemic as they had done in pre-pandemic (e.g. The emergency way heavily avoided that some mild patients choose to keep away from hospitals). In summary, in this particular period, the quarantine and social isolation just like some degree of grouping which exclude other potentially confounding risk factors. It makes us have gained a deeper appreciation of obvious risk factors for ICH. In view of the above-mentioned facts, we wish to make the following proposals: 1) People should pay more attention to modifiable risk factors such as alcohol intake and hypertension to reduce the risk of ICH in the COVID-19 pandemic period. 2) Various health organizations and clinicians should develop recommendations for

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people on how to use a healthy lifestyle to cope with physical distancing and social isolation. 3) Governments should explore effective prevention and intervention measures to prevent crises. As for SAH, we think it is hard to acquire any important conclusions due to too little data and a huge impact of a single article. It needs more authoritative studies to explore the impact of COVID-19 pandemic on SAH.

Limitations

Limitations to this study are the lack of information about stroke severity and prognoses, which might have to some extent underestimated the impact of the pandemic for stroke. Moreover, the total number of admissions in the pandemic is not reported, so the prevalence of stroke is inconclusive. Although our meta-analysis indicates higher morbidity of HS and ICH, the reliability needs to be verified by further researches because of the small sample size. Despite this, this study still has notable clinical and public health implications. For example, our data may encourage people to conduct further research to prove the potential pathogenesis of ICH. On the other hand, our results could attract more attention to the influence of COVID-19, thus people can perfect isolation approaches and prevention measures.

Conclusions

The study identifies moderate reductions in stroke admissions during the COVID-19 pandemic. Studies have shown that alcohol abuse and stress are important causes of increasing ICH hospitalizations during the pandemic. All governments, healthcare institutions, clinicians and academics should aim at preventing potential crisis in the COVID pandemic, exploring more effective policies for prevention and intervention.

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Author Contributions: YJY conceived and designed the study and drafted the manuscript. YJY and YLN acquired the data and performed data extraction. YJY interpreted the data, and performed extensive research on the topic. YJY performed the statistical analysis and made all tables and figures. FBS, JZ and SH checked the data. PD funded the project. XHW supervised the project. All authors contributed to the accomplishment of the manuscript.

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Disclosures: None.

Conflicts of interest: The authors declare that they have no competing interests.

Data availability statement: Data are available upon reasonable request. The data are available from the corresponding author (cematianya@foxmail.com) upon request.

Ethics approval: Neither ethics approval nor participant consent was required as this study was based solely on the summary results of previously published articles. Individual patient data were not obtained or accessed.

Patient and public involvement: Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

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References

1. Huang C, Wang Y, Li X, Ren L, Zhao J, Hu Y, et al. Clinical features of patients infected with 2019 novel coronavirus in wuhan, china. *The Lancet*. 2020;395:497-506
2. Sohrabi C, Alsafi Z, O'Neill N, Khan M, Kerwan A, Al-Jabir A, et al. World health organization declares global emergency: A review of the 2019 novel coronavirus (covid-19). *International journal of surgery (London, England)*. 2020;76:71-76
3. Lipsitch M, Swerdlow DL, Finelli L. Defining the epidemiology of covid-19 - studies needed. *The New England journal of medicine*. 2020;382:1194-1196
4. World Health O. Coronavirus disease (covid-19): Weekly epidemiological, update 1, 17 august 2020. 2020
5. GBD, Stroke, Collaborators. Global, regional, and national burden of stroke, 1990-2016: A systematic analysis for the global burden of disease study 2016. *The Lancet. Neurology*. 2019
6. Feigin VL, Nguyen G, Cercy K, Johnson CO, Alam T, Parmar PG, et al. Global, regional, and country-specific lifetime risks of stroke, 1990 and 2016. *The New England journal of medicine*. 2018;379:2429-2437
7. O'Donnell MJ, Chin SL, Rangarajan S, Xavier D, Liu L, Zhang H, et al. Global and regional effects of potentially modifiable risk factors associated with acute stroke in 32 countries (interstroke): A case-control study. *Lancet (London, England)*. 2016;388:761-775
8. Krishnamurthi RV, Ikeda T, Feigin VL. Global, regional and country-specific burden of ischaemic stroke, intracerebral haemorrhage and subarachnoid haemorrhage: A systematic analysis of the global burden of disease study 2017. *Neuroepidemiology*. 2020;54:171-179
9. Pandian JD, Gall SL, Kate MP, Silva GS, Akinyemi RO, Ovbiagele BI, et al. Prevention of stroke: A global perspective. *Lancet (London, England)*. 2018;392:1269-1278
10. Kimura K. [diagnosis and treatment in acute stroke]. *Rinsho shinkeigaku = Clinical neurology*. 2010;50:783-786
11. Solomon MD, McNulty EJ, Rana JS, Leong TK, Lee C, Sung SH, et al. The covid-19 pandemic and the incidence of acute myocardial infarction. *The New England journal of medicine*. 2020;383:691-693
12. July J, Pranata R. Impact of the coronavirus disease pandemic on the number of strokes and mechanical thrombectomies: A systematic review and meta-analysis. *Journal of stroke and cerebrovascular diseases : the official journal of National Stroke Association*. 2020;29:105185
13. Markus HS, Brainin M. Covid-19 and stroke—a global world stroke organization perspective. *International Journal of Stroke*. 2020;15:361-364
14. Wang X, Dong Y, Qi X, Huang C, Hou L. Cholesterol levels and risk of hemorrhagic stroke: A systematic review and meta-analysis. *Stroke*. 2013;44:1833-1839
15. Sacco S, Ricci S, Ornello R, Eusebi P, Petraglia L, Toni D. Reduced admissions for cerebrovascular events during covid-19 outbreak in italy. *Stroke*. 2020;51:3746-3750
16. Rameez F, McCarthy P, Cheng Y, Packard LM, Davis AT, Wees N, et al. Impact of a stay-at-home order on stroke admission, subtype, and metrics during the covid-19 pandemic. *Cerebrovascular diseases extra*. 2020;10:159-165
17. Balestrino M, Coccia A, Boffa AS, Furgani A, Bermano F, Finocchi C, et al. Request of hospital care dropped for tia but remained stable for stroke during covid-19 pandemic at a large italian university hospital. *Internal and emergency medicine*. 2020:1-5

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18. John S, Hussain SI, Piechowski-Jozwiak B, Dibu J, Kesav P, Bayrlee A, et al. Clinical characteristics and admission patterns of stroke patients during the covid 19 pandemic: A single center retrospective, observational study from the abu dhabi, united arab emirates. *Clinical neurology and neurosurgery*. 2020;199:106227
19. Luostarinen T, Virta J, Satopää J, Bäcklund M, Kivisaari R, Korja M, et al. Intensive care of traumatic brain injury and aneurysmal subarachnoid hemorrhage in helsinki during the covid-19 pandemic. *Acta neurochirurgica*. 2020;162:2715-2724
20. Zini A, Romoli M, Gentile M, Migliaccio L, Picoco C, Dell'Arciprete O, et al. The stroke mothership model survived during covid-19 era: An observational single-center study in emilia-romagna, italy. *Neurological sciences : official journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology*. 2020;41:3395-3399
21. Schwarz V, Mahfoud F, Lauder L, Reith W, Behnke S, Smola S, et al. Decline of emergency admissions for cardiovascular and cerebrovascular events after the outbreak of covid-19. *Clinical research in cardiology : official journal of the German Cardiac Society*. 2020;109:1500-1506
22. Sarfo FS, Mensah NO, Opoku FA, Adusei-Mensah N, Ampofo M, Ovbiagele B. Covid-19 and stroke: Experience in a ghanaian healthcare system. *Journal of the neurological sciences*. 2020;416:117044
23. Roushdy TM, Nahas NME, Aref HM, Georgy SS, Zaki AS, Bedros RY, et al. Stroke in the time of coronavirus disease 2019: Experience of two university stroke centers in egypt. *Journal of stroke*. 2020;22:275-277
24. Jansen R, Lee JA-O, Turowski B, Kaschner M, Caspers J, Bernhard M, et al. Consequences of covid-19 pandemic lockdown on emergency and stroke care in a german tertiary stroke center.
25. Uphaus T, Gröschel S, Hayani E, Hahn M, Steffen F, Gröschel K. Stroke care within the covid-19 pandemic-increasing awareness of transient and mild stroke symptoms needed.
26. Altunışık E, Arık A. Decreased stroke applications during pandemic: Collateral effects of covid-19. *Turkish Journal of Neurology/Turk Noroloji Dergisi*. 2021;27
27. Carson L, Kui C, Smith G, Dixit AK. The effect of the 2019 novel coronavirus pandemic on stroke and tia patient admissions: Perspectives and risk factors. Lid - 10.3390/jcm10071357 [doi] lid - 1357.
28. Gattringer T, Fandler-Höfler S, Kneihsl M, Hofer E, Köle W, Schmidt R, et al. Hospital admissions of acute cerebrovascular diseases during and after the first wave of the covid-19 pandemic: A state-wide experience from austria.
29. Ghoreishi A, Arsang-Jang S, Sabaa-Ayoun Z, Yassi N, Sylaja PN, Akbari Y, et al. Stroke care trends during covid-19 pandemic in zanjan province, iran. From the cascade initiative: Statistical analysis plan and preliminary results.
30. Hasan AA-O, Das SC, Islam MS, Mansur M, Shawon MSR, Hassan R, et al. Impact of covid-19 on hospital admission of acute stroke patients in bangladesh.
31. Ramírez-Moreno JA-O, Portilla-Cuenca JC, Hariramani-Ramchandani R, Rebollo B, Bermejo Casado I, Macías-Sedas P, et al. Slump in hospital admissions for stroke, a fact of an uncertain nature that requires explanation. Lid - 10.3390/brainsci11010092 [doi] lid - 92.
32. Velilla-Alonso G, García-Pastor A, Rodríguez-López Á, Gómez-Roldós A, Sánchez-Soblechero A, Amaya-Pascasio L, et al. Acute stroke care during the covid-19 pandemic: Reduction in the number of admissions of elderly patients and increase in prehospital delays.

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33. Ariesen MJ, Claus SP, Rinkel GJ, Algra A. Risk factors for intracerebral hemorrhage in the general population: A systematic review. *Stroke*. 2003;34:2060-2065
34. Ćosić K, Popović S, Šarlija M, Kesedžić I. Impact of human disasters and covid-19 pandemic on mental health: Potential of digital psychiatry. *Psychiatria Danubina*. 2020;32:25-31
35. Sun Y, Li Y, Bao Y, Meng S, Sun Y, Schumann G, et al. Brief report: Increased addictive internet and substance use behavior during the covid-19 pandemic in china. *The American journal on addictions*. 2020;29:268-270
36. Colbert S, Wilkinson C, Thornton L, Richmond R. Covid-19 and alcohol in australia: Industry changes and public health impacts. *Drug and alcohol review*. 2020;39:435-440
37. Kmiec J. President's message: The impact of the covid-19 public health emergency on the practice of addiction medicine. *Journal of addictive diseases*. 2020;38:235-236
38. Andrade C. Covid-19 and lockdown: Delayed effects on health. *Indian journal of psychiatry*. 2020;62:247-249
39. Rehm J, Kilian C, Ferreira-Borges C, Jernigan D, Monteiro M, Parry CDH, et al. Alcohol use in times of the covid 19: Implications for monitoring and policy. *Drug and alcohol review*. 2020;39:301-304
40. Carrico AW, Horvath KJ, Grov C, Moskowitz JT, Pahwa S, Pallikkuth S, et al. Double jeopardy: Methamphetamine use and hiv as risk factors for covid-19. *AIDS and behavior*. 2020;24:3020-3023
41. Clay JM, Parker MO. Alcohol use and misuse during the covid-19 pandemic: A potential public health crisis? *The Lancet. Public health*. 2020;5:e259
42. Shigemura J, Ursano RJ, Morganstein JC, Kurosawa M, Benedek DM. Public responses to the novel 2019 coronavirus (2019-ncov) in japan: Mental health consequences and target populations. *Psychiatry and clinical neurosciences*. 2020;74:281-282
43. Wang Y, Di Y, Ye J, Wei W. Study on the public psychological states and its related factors during the outbreak of coronavirus disease 2019 (covid-19) in some regions of china. *Psychology, health & medicine*. 2021;26:13-22
44. Ramalho R. Alcohol consumption and alcohol-related problems during the covid-19 pandemic: A narrative review. *Australasian psychiatry : bulletin of Royal Australian and New Zealand College of Psychiatrists*. 2020;28:524-526

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Figure legends

Figure 1. Flow chart of study selection process

Figure 2. (A) Forest plot showing the admission of HS during the pandemic compared with the control period. (B) Forest plot showing the admission of ICH during the pandemic compared with the control period. (C) Forest plot showing the admission of SAH during the pandemic compared with the control period.

Figure 3. Funnel plot with the pooled estimate of the fixed-effects model. (A) The funnel plot of HS group. (B) The funnel plot of ICH group.

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Table 1: Summary of characteristics of included studies.

Author	Study Design	Region	Age(years)	Male(%)	Pandemic Period	Control Period	Total number of Intracerebral hemorrhage admissions (Pandemic vs Control)	Total number of subarachnoid hemorrhage admissions (Pandemic vs Control)	Total number of hemorrhagic stroke admissions (Pandemic vs Control)	Total number of stroke admissions (Pandemic vs Control)
Balestrino et al, 2020	Single Center; Retrospective Observational	Italy	N/A*	N/A*	8st March 2020 to 2st May 2020	8st March 2019 to 2st May 2019	N/A*	N/A*	20 vs 17	124 vs 165
John et al, 2020	Single Center; Retrospective Observational	United Arab Emirates	48.9(±14.7) vs 49.3(±14.1)	66.7% vs 80.8%	1st March 2020 to10st May 2020	1st March 2019 to10st May 2019	24 vs 12	13 vs 13	42 vs 26	210 vs 148
Rameez et al, 2020	Single Center; Retrospective Observational	USA	65.6 vs 72.4	61.7% vs 51.3%	24st March 2020 to 23st April 2020	24st March 2019 to 23st April 2019	9 vs 12	0 vs 0	9 vs 12	47 vs 80
Zini et al, 2020	Single Center; Retrospective Observational	Italy	71.7(±18) vs 75.2(±13.8）	53.8% vs 50.7%	1st March 2020 to 30st April 2020	1st March 2019 to 30st April 2019	17 vs 21	N/A*	17 vs 21	145 vs 138
Schwarz et al, 2020	Single Center; Retrospective Observational	Germany	N/A*	N/A*	Calendar weeks 1 to16 in 2020	Calendar weeks 1 to16 in 2019	32 vs 24	N/A*	32 vs 24	286 vs 386

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Sarfo et al, 2020	Single Center; Retrospective Observational	Ghana	60.6(±14.8) vs 59.7(±15.4）	57.1% vs 53.4%	19st January 2020 to 20st June 2020	19st January 2019 to 20st June 2019	N/A*	N/A*	118 vs 111	431 vs 401
Sacco et al, 2020	Multicenter; Prospective and Retrospective	Italy	73.5(±12.9) vs 73.6(±13.3）	53% vs 53.4%	1st March 2020 to 31st March 2020	1st March 2019 to 31st March 2019	N/A*	N/A*	322 vs 400	2379 vs 3145
Roushdy et al, 2020	Single Center; Retrospective Observational	Egypt	63.2(±12.7) vs 63(±13.8）	61.3% vs 57%	15st February 2020 to 5st April 2020	15st February 2019 to 5st April 2019	7 vs 5	1 vs 3	8 vs 8	93 vs 151
Altunisik et al, 2021	Single Center; Retrospective Observational	Turkey	71.95(±12.37) vs 70.96(±13.52)	61.36% vs 53.93%	1st April 2020 to 31st May 2020	1st April 2019 to 31st May 2019	3 vs 8	N/A*	3 vs 8	40 vs 72
Luke et al, 2021	Single Center; Retrospective Observational	Switzerland	N/A*	N/A*	11st March 2020 to 26st May 2020	11st March 2019 to 26st May 2019	26 vs 19	N/A*	26 vs 19	200 vs 223
Thomas et al, 2021	Multicenter; Retrospective Observational	Austria	N/A*	N/A*	1st March 2020 to 31st May 2020	1st March 2019 to 31st May 2019	N/A*	N/A*	188 vs 219	1014 vs 1185
Abdoreza et al, 2020	Single Center; Retrospective Observational	Iran	69.21(±13.88) vs 68.74(±14.49)	55.17% vs 55.21%	18st February 2020 to 18st July 2020	18st February 2019 to 18st July 2019	48 vs 50	N/A*	48 vs 50	229 vs 352

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Hasan et al, 2021	Single Center; Retrospective Observational	Bangladesh	N/A*	N/A*	1st April 2020 to 30st June 2020	1st January 2020 to 31st March 2020	282 vs 449	52 vs 182	334 vs 631	487 vs 907
Robin et al, 2021	Single Center; Retrospective Observational	Germany	77(64-82) vs 79(69-85)	47.6% vs 45.7%	16st March 2020 to 12st April 2020	16st March 2019 to 12st April 2019	6 vs 6	N/A*	6 vs 6	63 vs 70
Ramírez-Moreno et al, 2021	Multicenter; Retrospective Observational	Spain	70.9(±2.9)	N/A*	1st January 2020 to 31st May 2020	1st January 2019 to 31st May 2019	N/A*	N/A*	44 vs 48	273 vs 451
Timo et al, 2020	Single Center; Retrospective Observational	Germany	72.9(±12.5) vs 72.6(±13.4)	46.6% vs 54.3%	March/April 2020	March/April 2019	7 vs 12	4 vs 4	9 vs 13	148 vs 207
Gabriel et al, 2021	Single Center; Retrospective Observational	Spain	68(53-79) vs 76(60.2-83)	51.8% vs 55.4%	14st March 2020 to 14st May 2020	14st March 2019 to 14st May 2019	N/A*	N/A*	7 vs 8	83 vs 112

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Table 2: Newcastle-Ottawa quality assessment scale and total score of each study.

Included Studies	Newcastle-Ottawa quality assessment scale								Score
	1	2	3	4	5	6	7	8	Total of number of stars
Balestrino et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
John et al, 2020	A*	A*	B	A*	A*	A*	A*	A*	7
Rameez et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Zini et al, 2020	A*	A*	B	A*	A*	A*	A*	A*	7
Schwarz et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Sarfo et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Sacco et al, 2020	A*	A*	A*	A*	A**	A*	A*	A*	9
Roushdy et al, 2020	B	B	B	A*	A*	A*	A*	A*	5
Altunisik et al, 2021	A*	A*	B	A*	A*	A*	A*	A*	7
Luke et al, 2021	A*	A*	B	A*	A*	A*	A*	A*	7
Thomas et al, 2021	A*	A*	A*	A*	A*	A*	A*	A*	8
Abdoreza et al, 2020	B	A*	B	A*	A*	A*	A*	A*	6
Hasan et al, 2021	A*	B	A*	A*	B	A*	A*	A*	6
Robin et al, 2021	A*	A*	A*	A*	A*	A*	A*	A*	8
Ramírez-Moreno et al, 2021	A*	A*	B	A*	A*	A*	A*	A*	7
Timo et al, 2020	A*	A*	A*	A*	A*	A*	A*	A*	8
Gabriel et al, 2021	A*	A*	A*	A*	A**	A*	A*	A*	9

Note: A*, one star; A, two stars; B, no star; 1, Adequate Case Definition; 2, Representativeness of Cases; 3, Selection of Controls; 4, Definition of Controls; 5, Comparability; 6, Ascertainment of Exposure; 7, Same Method; 8, Non-Response Rate.**

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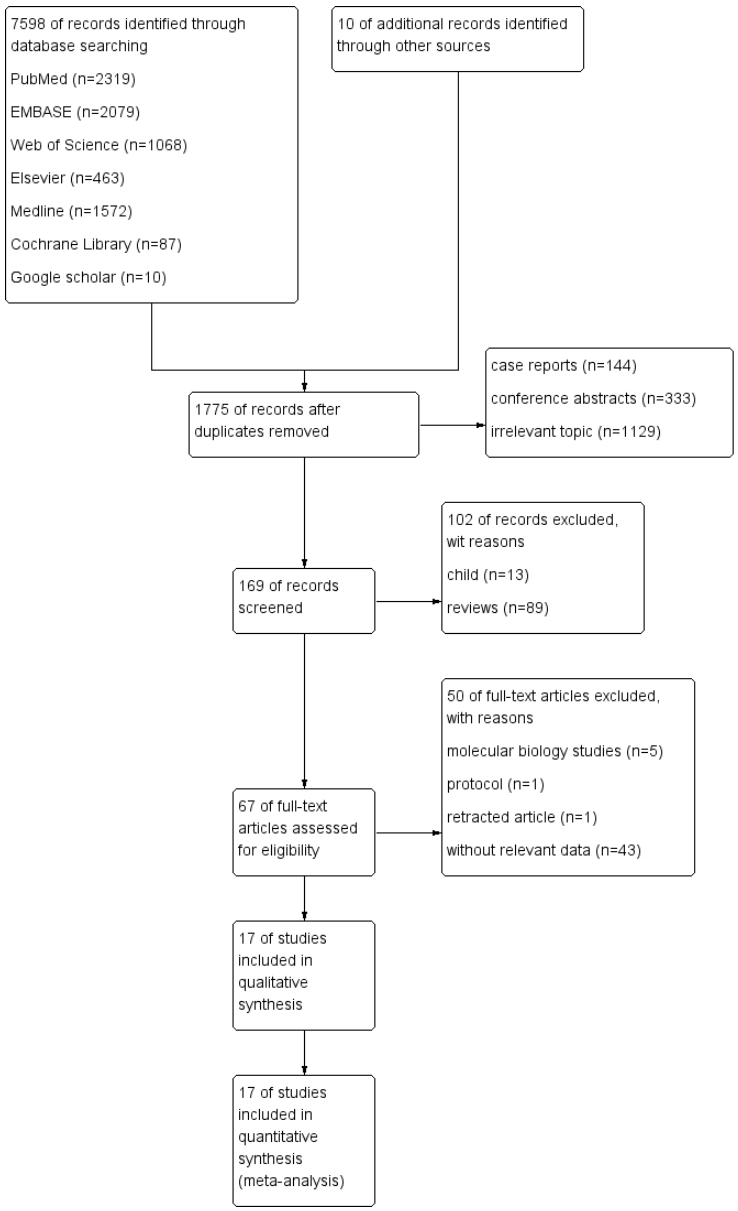


Figure 1. Flow chart of study selection process

234x384mm (72 x 72 DPI)

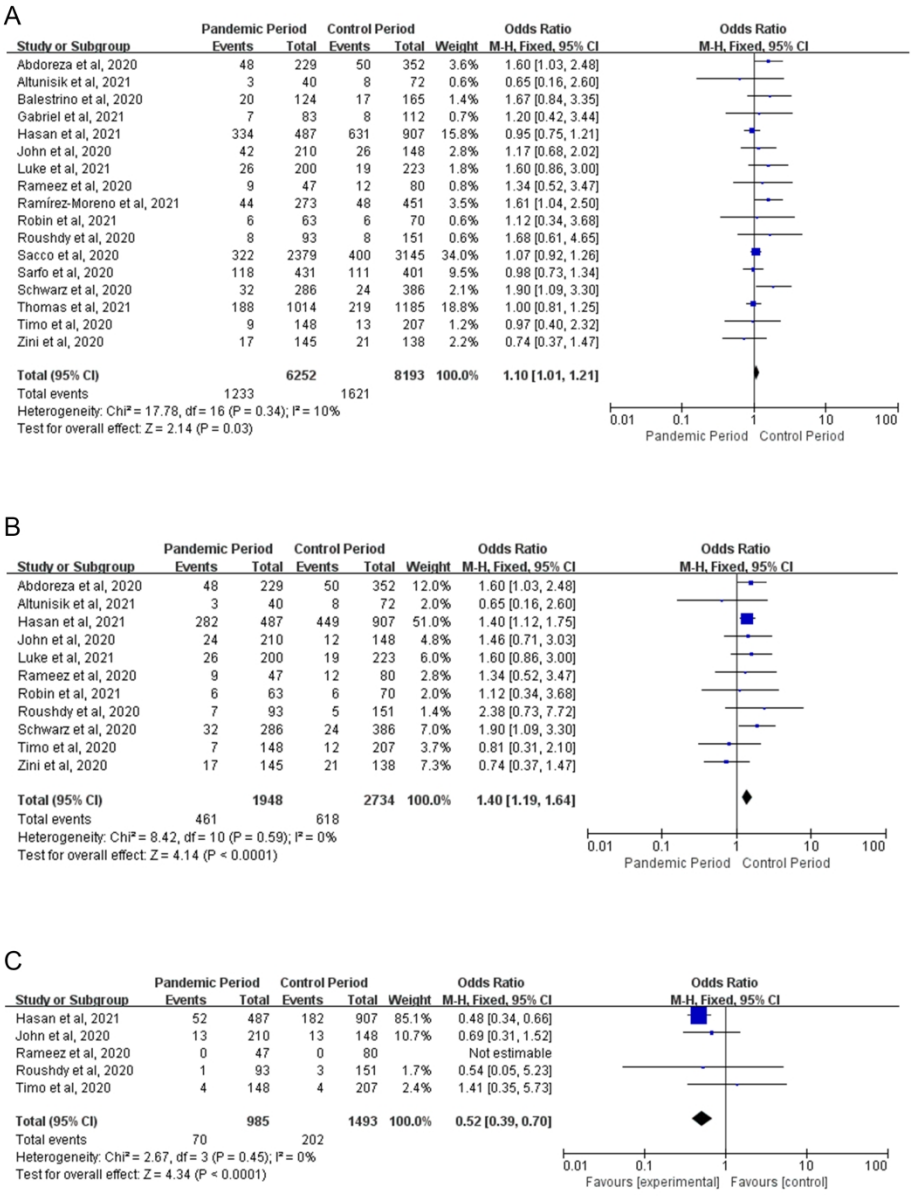


Figure 2. (A) Forest plot showing the admission of HS during the pandemic compared with the control period. (B) Forest plot showing the admission of ICH during the pandemic compared with the control period. (C) Forest plot showing the admission of SAH during the pandemic compared with the control period.

302x391mm (300 x 300 DPI)

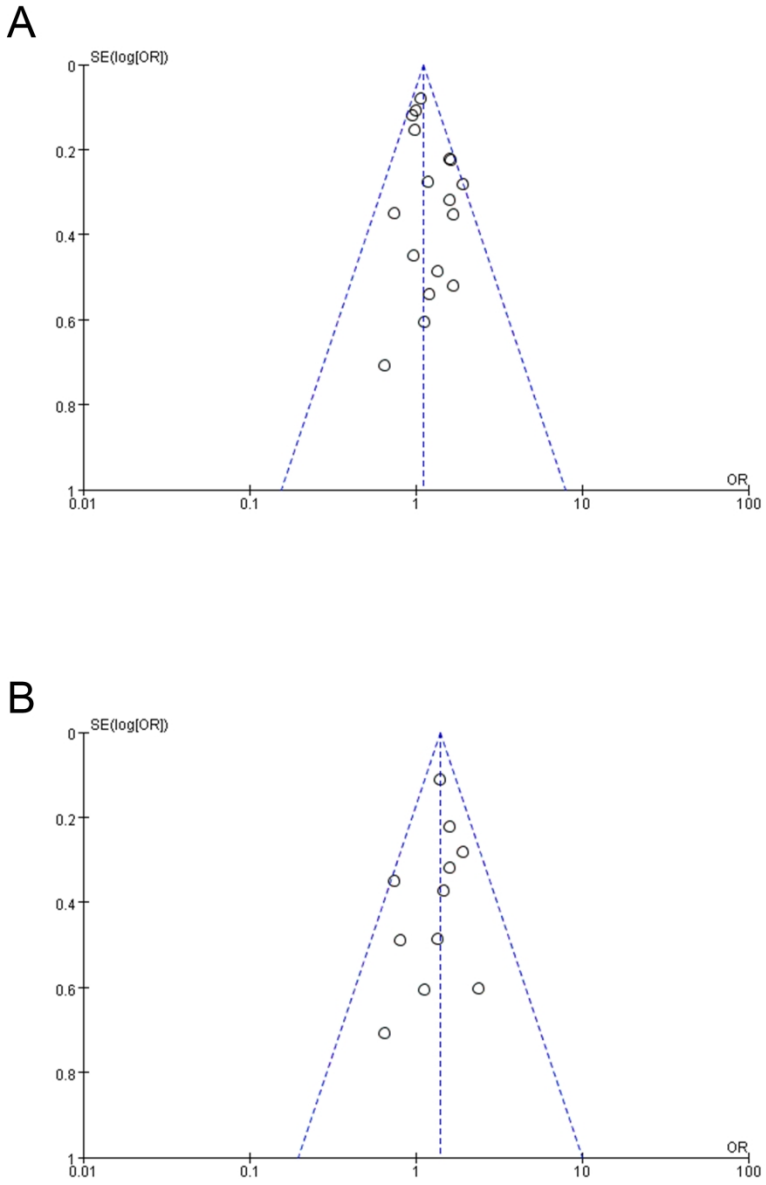


Figure 3. Funnel plot with the pooled estimate of the fixed-effects model. (A) The funnel plot of HS group. (B) The funnel plot of ICH group.

228x350mm (300 x 300 DPI)

Appendix 1: Search terms and strategies

EMBASE search strategy	
#7	#4 AND #6
#6	#1 OR #2 OR #3 OR #5
#5	'cerebral hemorrhage':ab,ti OR 'hemorrhage, cerebrum':ab,ti OR 'cerebrum hemorrhage':ab,ti OR 'cerebrum hemorrhages':ab,ti OR 'hemorrhages, cerebrum':ab,ti OR 'cerebral parenchymal hemorrhage':ab,ti OR 'cerebral parenchymal hemorrhages':ab,ti OR 'hemorrhage, cerebral parenchymal':ab,ti OR 'hemorrhages, cerebral parenchymal':ab,ti OR 'parenchymal hemorrhage, cerebral':ab,ti OR 'parenchymal hemorrhages, cerebral':ab,ti OR 'intracerebral hemorrhage':ab,ti OR 'hemorrhage, intracerebral':ab,ti OR 'hemorrhages, intracerebral':ab,ti OR 'intracerebral hemorrhages':ab,ti OR 'hemorrhage, cerebral':ab,ti OR 'cerebral hemorrhages':ab,ti OR 'hemorrhages, cerebral':ab,ti OR 'brain hemorrhage, cerebral':ab,ti OR 'brain hemorrhages, cerebral':ab,ti OR 'cerebral brain hemorrhage':ab,ti OR 'cerebral brain hemorrhages':ab,ti OR 'hemorrhage, cerebral brain':ab,ti OR 'hemorrhages, cerebral brain':ab,ti
#4	'covid-19':ab,ti OR 'covid 19':ab,ti OR 'covid-19 virus disease':ab,ti OR 'covid 19 virus disease':ab,ti OR 'covid-19 virus diseases':ab,ti OR 'disease, covid-19 virus':ab,ti OR 'virus disease, covid-19':ab,ti OR 'covid-19 virus infection':ab,ti OR 'covid 19 virus infection':ab,ti OR 'covid-19 virus infections':ab,ti OR 'infection, covid-19 virus':ab,ti OR 'virus infection, covid-19':ab,ti OR '2019-ncov infection':ab,ti OR '2019 ncov infection':ab,ti OR '2019-ncov infections':ab,ti OR 'infection, 2019-ncov':ab,ti OR 'coronavirus disease-19':ab,ti OR 'coronavirus disease 19':ab,ti OR '2019 novel coronavirus disease':ab,ti OR '2019 novel coronavirus infection':ab,ti OR '2019-ncov disease':ab,ti OR '2019 ncov disease':ab,ti OR '2019-ncov diseases':ab,ti OR 'disease, 2019-ncov':ab,ti OR 'covid19':ab,ti OR 'disease 2019, coronavirus':ab,ti OR 'sars coronavirus 2 infection':ab,ti OR 'sars-cov-2 infection':ab,ti OR 'infection, sars-cov-2':ab,ti OR 'sars cov 2 infection':ab,ti OR 'sars-cov-2 infections':ab,ti OR 'covid 19 pandemic':ab,ti OR 'covid-19 pandemics':ab,ti OR 'pandemic, covid-19':ab,ti
#3	'stroke':ab,ti OR 'strokes':ab,ti OR 'cerebrovascular accident':ab,ti OR 'cerebrovascular accidents':ab,ti OR 'cva (cerebrovascular accident)':ab,ti OR 'cvas (cerebrovascular accident)':ab,ti OR 'cerebrovascular apoplexy':ab,ti OR 'apoplexy, cerebrovascular':ab,ti OR 'vascular accident, brain':ab,ti OR 'brain vascular accident':ab,ti OR 'brain vascular accidents':ab,ti OR 'vascular accidents, brain':ab,ti OR 'cerebrovascular stroke':ab,ti OR 'cerebrovascular strokes':ab,ti OR 'stroke, cerebrovascular':ab,ti OR 'strokes, cerebrovascular':ab,ti OR 'apoplexy':ab,ti OR 'cerebral stroke':ab,ti OR 'cerebral strokes':ab,ti OR 'stroke, cerebral':ab,ti OR 'strokes, cerebral':ab,ti OR 'stroke, acute':ab,ti OR 'acute stroke':ab,ti OR 'acute strokes':ab,ti OR 'strokes, acute':ab,ti OR 'acute cerebrovascular accident':ab,ti OR 'acute cerebrovascular accidents':ab,ti OR 'cerebrovascular accidents, acute':ab,ti
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#1	'hemorrhagic stroke':ab,ti OR 'hemorrhagic strokes':ab,ti OR 'stroke, hemorrhagic':ab,ti OR 'subarachnoid hemorrhagic stroke':ab,ti OR 'hemorrhagic stroke, subarachnoid':ab,ti OR 'stroke, subarachnoid hemorrhagic':ab,ti OR 'subarachnoid hemorrhagic strokes':ab,ti OR 'intracerebral hemorrhagic stroke':ab,ti OR 'hemorrhagic stroke, intracerebral':ab,ti OR 'intracerebral hemorrhagic strokes':ab,ti OR 'stroke, intracerebral hemorrhagic':ab,ti OR 'intracerebral hemorrhage stroke':ab,ti OR 'hemorrhage stroke, intracerebral':ab,ti OR 'intracerebral hemorrhage strokes':ab,ti OR 'stroke, intracerebral hemorrhage':ab,ti
PubMed search strategy	
#17	((COVID-19[MeSH Major Topic]) OR (COVID 19[Title/Abstract] or COVID-19 Virus Disease[Title/Abstract] or COVID 19 Virus Disease[Title/Abstract] or COVID-19 Virus Diseases[Title/Abstract] or Disease, COVID-19 Virus[Title/Abstract] or Virus Disease, COVID-19[Title/Abstract] or COVID-19 Virus Infection[Title/Abstract] or COVID 19 Virus Infection[Title/Abstract] or COVID-19 Virus Infections[Title/Abstract] or Infection, COVID-19 Virus[Title/Abstract] or Virus Infection, COVID-19[Title/Abstract] or 2019-nCoV Infection[Title/Abstract] or 2019 nCoV Infection[Title/Abstract] or 2019-nCoV Infections[Title/Abstract] or Infection, 2019-nCoV[Title/Abstract] or Coronavirus Disease-19[Title/Abstract] or Coronavirus Disease 19[Title/Abstract] or 2019 Novel Coronavirus Disease[Title/Abstract] or 2019 Novel Coronavirus Infection[Title/Abstract] or 2019-nCoV Disease[Title/Abstract] or 2019 nCoV Disease[Title/Abstract] or 2019-nCoV Diseases[Title/Abstract] or Disease, 2019-nCoV[Title/Abstract] or COVID19[Title/Abstract] or Coronavirus Disease 2019[Title/Abstract] or Disease 2019, Coronavirus[Title/Abstract] or SARS Coronavirus 2 Infection[Title/Abstract] or SARS-CoV-2 Infection[Title/Abstract] or Infection, SARS-CoV-2[Title/Abstract] or SARS CoV 2 Infection[Title/Abstract] or COVID-19 Pandemic[Title/Abstract] or SARS-CoV-2 Infections[Title/Abstract] or COVID 19 Pandemic[Title/Abstract] or COVID-19 Pandemics[Title/Abstract] or Pandemic, COVID-19[Title/Abstract])) AND (((((cerebral hemorrhage[MeSH Major Topic]) OR (Hemorrhage, Cerebrum[Title/Abstract] or Cerebrum Hemorrhage[Title/Abstract] or Cerebrum Hemorrhages[Title/Abstract] or Hemorrhages, Cerebrum[Title/Abstract] or Cerebral Parenchymal Hemorrhage[Title/Abstract] or Cerebral Parenchymal Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Parenchymal[Title/Abstract] or Hemorrhages, Cerebral Parenchymal[Title/Abstract] or Parenchymal Hemorrhage, Cerebral[Title/Abstract] or Parenchymal Hemorrhages, Cerebral[Title/Abstract] or Intracerebral Hemorrhage[Title/Abstract] or Hemorrhage, Intracerebral[Title/Abstract] or Hemorrhages, Intracerebral[Title/Abstract] or Intracerebral Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral[Title/Abstract] or Cerebral Hemorrhages[Title/Abstract] or Hemorrhages, Cerebral[Title/Abstract] or Brain Hemorrhage, Cerebral[Title/Abstract] or Brain Hemorrhages, Cerebral[Title/Abstract] or Cerebral Brain Hemorrhage[Title/Abstract] or Cerebral Brain Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Brain[Title/Abstract] or Hemorrhages, Cerebral Brain[Title/Abstract])) OR ((Stroke[MeSH Major Topic]) OR (Strokes[Title/Abstract] or Cerebrovascular Accident[Title/Abstract] or Cerebrovascular Accidents[Title/Abstract] or CVA (Cerebrovascular Accident)[Title/Abstract] or CVAs (Cerebrovascular Accident)[Title/Abstract] or Cerebrovascular Apoplexy[Title/Abstract] or Apoplexy, Cerebrovascular[Title/Abstract] or Vascular Accident, Brain[Title/Abstract] or Brain Vascular Accident[Title/Abstract] or Brain Vascular Accidents[Title/Abstract] or Vascular Accidents, Brain[Title/Abstract] or Cerebrovascular Stroke[Title/Abstract] or Cerebrovascular Strokes[Title/Abstract] or Stroke, Cerebrovascular[Title/Abstract] or Strokes, Cerebrovascular[Title/Abstract] or Apoplexy[Title/Abstract] or Cerebral Stroke[Title/Abstract] or Cerebral Strokes[Title/Abstract] or Stroke, Cerebral[Title/Abstract] or Strokes, Cerebral[Title/Abstract] or Stroke, Acute[Title/Abstract] or Acute Stroke[Title/Abstract] or Acute Strokes[Title/Abstract] or Strokes, Acute[Title/Abstract] or Cerebrovascular Accident, Acute[Title/Abstract] or Acute Cerebrovascular Accident[Title/Abstract] or Acute Cerebrovascular Accidents[Title/Abstract] or Cerebrovascular Accidents, Acute[Title/Abstract]))) OR ((subarachnoid hemorrhage[MeSH Major Topic]) OR (SAH [Title/Abstract] or SAHs[Title/Abstract] or Hemorrhage, Subarachnoid[Title/Abstract] or Hemorrhages, Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Aneurysmal[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhage[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhages[Title/Abstract] or Hemorrhage, Aneurysmal Subarachnoid[Title/Abstract] or Hemorrhages, Aneurysmal Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages, Aneurysmal[Title/Abstract] or Subarachnoid Hemorrhage, Spontaneous[Title/Abstract] or Hemorrhage, Spontaneous Subarachnoid[Title/Abstract] or Hemorrhages, Spontaneous Subarachnoid[Title/Abstract] or Spontaneous Subarachnoid Hemorrhage[Title/Abstract] or Spontaneous Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Spontaneous[Title/Abstract] or Perinatal Subarachnoid Hemorrhage[Title/Abstract] or Hemorrhage, Perinatal Subarachnoid[Title/Abstract] or Hemorrhages, Perinatal Subarachnoid[Title/Abstract] or Perinatal Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Perinatal[Title/Abstract] or Subarachnoid Hemorrhages, Perinatal[Title/Abstract] or Subarachnoid Hemorrhage, Intracranial[Title/Abstract] or Hemorrhage, Intracranial Subarachnoid[Title/Abstract] or Hemorrhages, Intracranial Subarachnoid[Title/Abstract] or Intracranial Subarachnoid Hemorrhage[Title/Abstract] or Intracranial Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Intracranial[Title/Abstract]))) OR ((hemorrhagic stroke[MeSH Major Topic]) OR (Hemorrhagic Strokes[Title/Abstract] or Stroke, Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Subarachnoid[Title/Abstract] or Stroke, Subarachnoid Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Strokes[Title/Abstract] or Intracerebral Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhagic Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhagic[Title/Abstract] or Intracerebral Hemorrhage Stroke[Title/Abstract] or Hemorrhage Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhage Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhage[Title/Abstract])))
#16	(COVID-19[MeSH Major Topic]) OR (COVID 19[Title/Abstract] or COVID-19 Virus Disease[Title/Abstract] or COVID 19 Virus Disease[Title/Abstract] or COVID-19 Virus Diseases[Title/Abstract] or Disease, COVID-19 Virus[Title/Abstract] or Virus Disease, COVID-19[Title/Abstract] or COVID-19 Virus Infection[Title/Abstract] or COVID 19 Virus Infection[Title/Abstract] or COVID-19 Virus Infections[Title/Abstract] or Infection, COVID-19 Virus[Title/Abstract] or Virus Infection, COVID-19[Title/Abstract] or 2019-nCoV Infection[Title/Abstract] or 2019 nCoV Infection[Title/Abstract] or 2019-nCoV Infections[Title/Abstract] or Infection, 2019-nCoV[Title/Abstract] or Coronavirus Disease-19[Title/Abstract] or Coronavirus Disease 19[Title/Abstract] or 2019 Novel Coronavirus Disease[Title/Abstract] or 2019 Novel Coronavirus Infection[Title/Abstract] or 2019-nCoV Disease[Title/Abstract] or 2019 nCoV Disease[Title/Abstract] or 2019-nCoV Diseases[Title/Abstract] or Disease, 2019-nCoV[Title/Abstract] or COVID19[Title/Abstract] or Coronavirus Disease 2019[Title/Abstract] or Disease 2019, Coronavirus[Title/Abstract] or SARS Coronavirus 2 Infection[Title/Abstract] or SARS-CoV-2 Infection[Title/Abstract] or Infection, SARS-CoV-2[Title/Abstract] or SARS CoV 2 Infection[Title/Abstract] or COVID-19 Pandemic[Title/Abstract] or SARS-CoV-2 Infections[Title/Abstract] or COVID 19 Pandemic[Title/Abstract] or COVID-19 Pandemics[Title/Abstract] or Pandemic, COVID-19[Title/Abstract])
#15	COVID 19[Title/Abstract] or COVID-19 Virus Disease[Title/Abstract] or COVID 19 Virus Disease[Title/Abstract] or COVID-19 Virus Diseases[Title/Abstract] or Disease, COVID-19 Virus[Title/Abstract] or Virus Disease, COVID-19[Title/Abstract] or COVID-19 Virus Infection[Title/Abstract] or COVID 19 Virus Infection[Title/Abstract] or COVID-19 Virus Infections[Title/Abstract] or Infection, COVID-19 Virus[Title/Abstract] or Virus Infection, COVID-19[Title/Abstract] or 2019-nCoV Infection[Title/Abstract] or 2019 nCoV Infection[Title/Abstract] or 2019-nCoV Infections[Title/Abstract] or Infection, 2019-nCoV[Title/Abstract] or Coronavirus Disease-19[Title/Abstract] or Coronavirus Disease 19[Title/Abstract] or 2019 Novel Coronavirus Disease[Title/Abstract] or 2019 Novel Coronavirus Infection[Title/Abstract] or 2019-nCoV Disease[Title/Abstract] or 2019 nCoV Disease[Title/Abstract] or 2019-nCoV Diseases[Title/Abstract] or Disease, 2019-nCoV[Title/Abstract] or COVID19[Title/Abstract] or Coronavirus Disease 2019[Title/Abstract] or Disease 2019, Coronavirus[Title/Abstract] or SARS Coronavirus 2 Infection[Title/Abstract] or SARS-CoV-2 Infection[Title/Abstract] or Infection, SARS-CoV-2[Title/Abstract] or SARS CoV 2 Infection[Title/Abstract] or COVID-19 Pandemic[Title/Abstract] or SARS-CoV-2 Infections[Title/Abstract] or COVID 19 Pandemic[Title/Abstract] or COVID-19 Pandemics[Title/Abstract] or Pandemic, COVID-19[Title/Abstract]

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#14	COVID-19[MeSH Major Topic]
#13	((((cerebral hemorrhage[MeSH Major Topic]) OR (Hemorrhage, Cerebrum[Title/Abstract] or Cerebrum Hemorrhage[Title/Abstract] or Cerebrum Hemorrhages[Title/Abstract] or Hemorrhages, Cerebrum[Title/Abstract] or Cerebral Parenchymal Hemorrhage[Title/Abstract] or Cerebral Parenchymal Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Parenchymal[Title/Abstract] or Hemorrhages, Cerebral Parenchymal[Title/Abstract] or Parenchymal Hemorrhage, Cerebral[Title/Abstract] or Parenchymal Hemorrhages, Cerebral[Title/Abstract] or Intracerebral Hemorrhage[Title/Abstract] or Hemorrhage, Intracerebral[Title/Abstract] or Hemorrhages, Intracerebral[Title/Abstract] or Intracerebral Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral[Title/Abstract] or Cerebral Hemorrhages[Title/Abstract] or Hemorrhages, Cerebral[Title/Abstract] or Brain Hemorrhage, Cerebral[Title/Abstract] or Brain Hemorrhages, Cerebral[Title/Abstract] or Cerebral Brain Hemorrhage[Title/Abstract] or Cerebral Brain Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Brain[Title/Abstract] or Hemorrhages, Cerebral Brain[Title/Abstract])) OR ((Stroke[MeSH Major Topic]) OR (Strokes[Title/Abstract] or Cerebrovascular Accident[Title/Abstract] or Cerebrovascular Accidents[Title/Abstract] or CVA (Cerebrovascular Accident)[Title/Abstract] or CVAs (Cerebrovascular Accident)[Title/Abstract] or Cerebrovascular Apoplexy[Title/Abstract] or Apoplexy, Cerebrovascular[Title/Abstract] or Vascular Accident, Brain[Title/Abstract] or Brain Vascular Accident[Title/Abstract] or Brain Vascular Accidents[Title/Abstract] or Vascular Accidents, Brain[Title/Abstract] or Cerebrovascular Stroke[Title/Abstract] or Cerebrovascular Strokes[Title/Abstract] or Stroke, Cerebrovascular[Title/Abstract] or Strokes, Cerebrovascular[Title/Abstract] or Apoplexy[Title/Abstract] or Cerebral Stroke[Title/Abstract] or Cerebral Strokes[Title/Abstract] or Stroke, Cerebral[Title/Abstract] or Strokes, Cerebral[Title/Abstract] or Stroke, Acute[Title/Abstract] or Acute Stroke[Title/Abstract] or Acute Strokes[Title/Abstract] or Strokes, Acute[Title/Abstract] or Cerebrovascular Accident, Acute[Title/Abstract] or Acute Cerebrovascular Accident[Title/Abstract] or Acute Cerebrovascular Accidents[Title/Abstract] or Cerebrovascular Accidents, Acute[Title/Abstract]))) OR ((subarachnoid hemorrhage[MeSH Major Topic]) OR (SAH [Title/Abstract] or SAHs[Title/Abstract] or Hemorrhage, Subarachnoid[Title/Abstract] or Hemorrhages, Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Aneurysmal[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhage[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhages[Title/Abstract] or Hemorrhage, Aneurysmal Subarachnoid[Title/Abstract] or Hemorrhages, Aneurysmal Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages, Aneurysmal[Title/Abstract] or Subarachnoid Hemorrhage, Spontaneous[Title/Abstract] or Hemorrhage, Spontaneous Subarachnoid[Title/Abstract] or Hemorrhages, Spontaneous Subarachnoid[Title/Abstract] or Spontaneous Subarachnoid Hemorrhage[Title/Abstract] or Spontaneous Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Spontaneous[Title/Abstract] or Perinatal Subarachnoid Hemorrhage[Title/Abstract] or Hemorrhage, Perinatal Subarachnoid[Title/Abstract] or Hemorrhages, Perinatal Subarachnoid[Title/Abstract] or Perinatal Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Perinatal[Title/Abstract] or Subarachnoid Hemorrhages, Perinatal[Title/Abstract] or Subarachnoid Hemorrhage, Intracranial[Title/Abstract] or Hemorrhage, Intracranial Subarachnoid[Title/Abstract] or Hemorrhages, Intracranial Subarachnoid[Title/Abstract] or Intracranial Subarachnoid Hemorrhage[Title/Abstract] or Intracranial Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Intracranial[Title/Abstract]))) OR ((hemorrhagic stroke[MeSH Major Topic]) OR (Hemorrhagic Strokes[Title/Abstract] or Stroke, Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Subarachnoid[Title/Abstract] or Stroke, Subarachnoid Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Strokes[Title/Abstract] or Intracerebral Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhagic Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhagic[Title/Abstract] or Intracerebral Hemorrhage Stroke[Title/Abstract] or Hemorrhage Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhage Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhage[Title/Abstract]))
#12	(hemorrhagic stroke[MeSH Major Topic]) OR (Hemorrhagic Strokes[Title/Abstract] or Stroke, Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Subarachnoid[Title/Abstract] or Stroke, Subarachnoid Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Strokes[Title/Abstract] or Intracerebral Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhagic Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhagic[Title/Abstract] or Intracerebral Hemorrhage Stroke[Title/Abstract] or Hemorrhage Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhage Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhage[Title/Abstract])
#11	Hemorrhagic Strokes[Title/Abstract] or Stroke, Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Subarachnoid[Title/Abstract] or Stroke, Subarachnoid Hemorrhagic[Title/Abstract] or Subarachnoid Hemorrhagic Strokes[Title/Abstract] or Intracerebral Hemorrhagic Stroke[Title/Abstract] or Hemorrhagic Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhagic Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhagic[Title/Abstract] or Intracerebral Hemorrhage Stroke[Title/Abstract] or Hemorrhage Stroke, Intracerebral[Title/Abstract] or Intracerebral Hemorrhage Strokes[Title/Abstract] or Stroke, Intracerebral Hemorrhage[Title/Abstract]
#10	hemorrhagic stroke[MeSH Major Topic]
#9	(subarachnoid hemorrhage[MeSH Major Topic]) OR (SAH [Title/Abstract] or SAHs[Title/Abstract] or Hemorrhage, Subarachnoid[Title/Abstract] or Hemorrhages, Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Aneurysmal[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhage[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhages[Title/Abstract] or Hemorrhage, Aneurysmal Subarachnoid[Title/Abstract] or Hemorrhages, Aneurysmal Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages, Aneurysmal[Title/Abstract] or Subarachnoid Hemorrhage, Spontaneous[Title/Abstract] or Hemorrhage, Spontaneous Subarachnoid[Title/Abstract] or Hemorrhages, Spontaneous Subarachnoid[Title/Abstract] or Spontaneous Subarachnoid Hemorrhage[Title/Abstract] or Spontaneous Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Spontaneous[Title/Abstract] or Perinatal Subarachnoid Hemorrhage[Title/Abstract] or Hemorrhage, Perinatal Subarachnoid[Title/Abstract] or Hemorrhages, Perinatal Subarachnoid[Title/Abstract] or Perinatal Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Perinatal[Title/Abstract] or Subarachnoid Hemorrhages, Perinatal[Title/Abstract] or Subarachnoid Hemorrhage, Intracranial[Title/Abstract] or Hemorrhage, Intracranial Subarachnoid[Title/Abstract] or Hemorrhages, Intracranial Subarachnoid[Title/Abstract] or Intracranial Subarachnoid Hemorrhage[Title/Abstract] or Intracranial Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Intracranial[Title/Abstract])
#8	SAH [Title/Abstract] or SAHs[Title/Abstract] or Hemorrhage, Subarachnoid[Title/Abstract] or Hemorrhages, Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Aneurysmal[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhage[Title/Abstract] or Aneurysmal Subarachnoid Hemorrhages[Title/Abstract] or Hemorrhage, Aneurysmal Subarachnoid[Title/Abstract] or Hemorrhages, Aneurysmal Subarachnoid[Title/Abstract] or Subarachnoid Hemorrhages, Aneurysmal[Title/Abstract] or Subarachnoid Hemorrhage, Spontaneous[Title/Abstract] or Hemorrhage, Spontaneous Subarachnoid[Title/Abstract] or Hemorrhages, Spontaneous Subarachnoid[Title/Abstract] or Spontaneous Subarachnoid Hemorrhage[Title/Abstract] or Spontaneous Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Spontaneous[Title/Abstract] or Perinatal Subarachnoid Hemorrhage[Title/Abstract] or Hemorrhage, Perinatal Subarachnoid[Title/Abstract] or Hemorrhages, Perinatal Subarachnoid[Title/Abstract] or Perinatal Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhage, Perinatal[Title/Abstract] or Subarachnoid Hemorrhages, Perinatal[Title/Abstract] or Subarachnoid Hemorrhage, Intracranial[Title/Abstract] or Hemorrhage, Intracranial Subarachnoid[Title/Abstract] or Hemorrhages, Intracranial Subarachnoid[Title/Abstract] or Intracranial Subarachnoid Hemorrhage[Title/Abstract] or Intracranial Subarachnoid Hemorrhages[Title/Abstract] or Subarachnoid Hemorrhages, Intracranial[Title/Abstract]
#7	subarachnoid hemorrhage[MeSH Major Topic]
#6	(Stroke[MeSH Major Topic]) OR (Strokes[Title/Abstract] or Cerebrovascular Accident[Title/Abstract] or Cerebrovascular Accidents[Title/Abstract] or CVA (Cerebrovascular Accident)[Title/Abstract] or CVAs (Cerebrovascular Accident)[Title/Abstract] or Cerebrovascular Apoplexy[Title/Abstract] or Apoplexy, Cerebrovascular[Title/Abstract] or Vascular Accident, Brain[Title/Abstract] or Brain Vascular Accident[Title/Abstract] or Brain Vascular Accidents[Title/Abstract] or Vascular Accidents, Brain[Title/Abstract] or Cerebrovascular Stroke[Title/Abstract] or Cerebrovascular Strokes[Title/Abstract] or Stroke, Cerebrovascular[Title/Abstract] or Strokes, Cerebrovascular[Title/Abstract] or Apoplexy[Title/Abstract] or Cerebral Stroke[Title/Abstract] or Cerebral Strokes[Title/Abstract] or Stroke, Cerebral[Title/Abstract] or Strokes, Cerebral[Title/Abstract] or Stroke, Acute[Title/Abstract] or Acute Stroke[Title/Abstract] or Acute Strokes[Title/Abstract] or Strokes, Acute[Title/Abstract] or Cerebrovascular Accident, Acute[Title/Abstract] or Acute Cerebrovascular Accident[Title/Abstract] or Acute Cerebrovascular Accidents[Title/Abstract] or Cerebrovascular Accidents, Acute[Title/Abstract])
#5	Strokes[Title/Abstract] or Cerebrovascular Accident[Title/Abstract] or Cerebrovascular Accidents[Title/Abstract] or CVA (Cerebrovascular Accident)[Title/Abstract] or CVAs (Cerebrovascular Accident)[Title/Abstract] or Cerebrovascular Apoplexy[Title/Abstract] or Apoplexy, Cerebrovascular[Title/Abstract] or Vascular Accident, Brain[Title/Abstract] or Brain Vascular Accident[Title/Abstract] or Brain Vascular Accidents[Title/Abstract] or Vascular Accidents, Brain[Title/Abstract] or Cerebrovascular Stroke[Title/Abstract] or Cerebrovascular Strokes[Title/Abstract] or Stroke, Cerebrovascular[Title/Abstract] or Strokes, Cerebrovascular[Title/Abstract] or Apoplexy[Title/Abstract] or Cerebral Stroke[Title/Abstract] or Cerebral Strokes[Title/Abstract] or Stroke, Cerebral[Title/Abstract] or Strokes, Cerebral[Title/Abstract] or Stroke, Acute[Title/Abstract] or Acute Stroke[Title/Abstract] or Acute Strokes[Title/Abstract] or Strokes, Acute[Title/Abstract] or Cerebrovascular Accident, Acute[Title/Abstract] or Acute Cerebrovascular Accident[Title/Abstract] or Acute Cerebrovascular Accidents[Title/Abstract] or Cerebrovascular Accidents, Acute[Title/Abstract]
#4	Stroke[MeSH Major Topic]
#3	(cerebral hemorrhage[MeSH Major Topic]) OR (Hemorrhage, Cerebrum[Title/Abstract] or Cerebrum Hemorrhage[Title/Abstract] or Cerebrum Hemorrhages[Title/Abstract] or Hemorrhages, Cerebrum[Title/Abstract] or Cerebral Parenchymal Hemorrhage[Title/Abstract] or Cerebral Parenchymal Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Parenchymal[Title/Abstract] or Hemorrhages, Cerebral Parenchymal[Title/Abstract] or Parenchymal Hemorrhage, Cerebral[Title/Abstract] or Parenchymal Hemorrhages, Cerebral[Title/Abstract] or Intracerebral Hemorrhage[Title/Abstract] or Hemorrhage, Intracerebral[Title/Abstract] or Hemorrhages, Intracerebral[Title/Abstract] or Intracerebral Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral[Title/Abstract] or Cerebral Hemorrhages[Title/Abstract] or Hemorrhages, Cerebral[Title/Abstract] or Brain Hemorrhage, Cerebral[Title/Abstract] or Brain Hemorrhages, Cerebral[Title/Abstract] or Cerebral Brain Hemorrhage[Title/Abstract] or Cerebral Brain Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Brain[Title/Abstract] or Hemorrhages, Cerebral Brain[Title/Abstract])
#2	Hemorrhage, Cerebrum[Title/Abstract] or Cerebrum Hemorrhage[Title/Abstract] or Cerebrum Hemorrhages[Title/Abstract] or Hemorrhages, Cerebrum[Title/Abstract] or Cerebral Parenchymal Hemorrhage[Title/Abstract] or Cerebral Parenchymal Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Parenchymal[Title/Abstract] or Hemorrhages, Cerebral Parenchymal[Title/Abstract] or Parenchymal Hemorrhage, Cerebral[Title/Abstract] or Parenchymal Hemorrhages, Cerebral[Title/Abstract] or Intracerebral Hemorrhage[Title/Abstract] or Hemorrhage, Intracerebral[Title/Abstract] or Hemorrhages, Intracerebral[Title/Abstract] or Intracerebral Hemorrhages[Title/Abstract] or Hemorrhages, Cerebral[Title/Abstract] or Brain Hemorrhage, Cerebral[Title/Abstract] or Brain Hemorrhages, Cerebral[Title/Abstract] or Cerebral Brain Hemorrhage[Title/Abstract] or Cerebral Brain Hemorrhages[Title/Abstract] or Hemorrhage, Cerebral Brain[Title/Abstract] or Hemorrhages, Cerebral Brain[Title/Abstract]

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#1	cerebral hemorrhage[MeSH Major Topic]
Web of Science search strategy	
#7	(#6) AND #2
#6	(((#5) OR #4) OR #3) OR #1
#5	(TI=(hemorrhagic stroke)) or TI=(Hemorrhagic Strokes) or TI=(Stroke, Hemorrhagic) or TI=(Subarachnoid Hemorrhagic Stroke) or TI=(Hemorrhagic Stroke, Subarachnoid) or TI=(Stroke, Subarachnoid Hemorrhagic) or TI=(Subarachnoid Hemorrhagic Strokes) or TI=(Intracerebral Hemorrhagic Stroke) or TI=(Hemorrhagic Stroke, Intracerebral) or TI=(Intracerebral Hemorrhagic Strokes) or TI=(Stroke, Intracerebral Hemorrhagic) or TI=(Intracerebral Hemorrhage Stroke) or TI=(Hemorrhage Stroke, Intracerebral) or TI=(Intracerebral Hemorrhage Strokes) or TI=(Stroke, Intracerebral Hemorrhage)
#4	(TI=(subarachnoid hemorrhage)) or TI=(SAH) or TI=(SAHs) or TI=(Hemorrhage, Subarachnoid) or TI=(Hemorrhages, Subarachnoid) or TI=(Subarachnoid Hemorrhages) or TI=(Subarachnoid Hemorrhage, Aneurysmal) or TI=(Aneurysmal Subarachnoid Hemorrhage) or TI=(Aneurysmal Subarachnoid Hemorrhages) or TI=(Hemorrhage, Aneurysmal Subarachnoid) or TI=(Hemorrhages, Aneurysmal Subarachnoid) or TI=(Subarachnoid Hemorrhages, Aneurysmal) or TI=(Subarachnoid Hemorrhage, Spontaneous) or TI=(Hemorrhage, Spontaneous Subarachnoid) or TI=(Hemorrhages, Spontaneous Subarachnoid) or TI=(Spontaneous Subarachnoid Hemorrhage) or TI=(Spontaneous Subarachnoid Hemorrhages) or TI=(Subarachnoid Hemorrhages, Spontaneous) or TI=(Perinatal Subarachnoid Hemorrhage) or TI=(Hemorrhage, Perinatal Subarachnoid) or TI=(Hemorrhages, Perinatal Subarachnoid) or TI=(Perinatal Subarachnoid Hemorrhages) or TI=(Subarachnoid Hemorrhage, Perinatal) or TI=(Subarachnoid Hemorrhages, Perinatal) or TI=(Subarachnoid Hemorrhage, Intracranial) or TI=(Hemorrhage, Intracranial Subarachnoid) or TI=(Hemorrhages, Intracranial Subarachnoid) or TI=(Intracranial Subarachnoid Hemorrhage) or TI=(Intracranial Subarachnoid Hemorrhages) or TI=(Subarachnoid Hemorrhages, Intracranial)
#3	(TI=(Stroke)) or TI=(Strokes) or TI=(Cerebrovascular Accident) or TI=(Cerebrovascular Accidents) or TI=(CVA (Cerebrovascular Accident)) or TI=(CVAs (Cerebrovascular Accident)) or TI=(Cerebrovascular Apoplexy) or TI=(Apoplexy, Cerebrovascular) or TI=(Vascular Accident, Brain) or TI=(Brain Vascular Accident) or TI=(Brain Vascular Accidents) or TI=(Vascular Accidents, Brain) or TI=(Cerebrovascular Stroke) or TI=(Cerebrovascular Strokes) or TI=(Stroke, Cerebrovascular) or TI=(Strokes, Cerebrovascular) or TI=(Apoplexy) or TI=(Cerebral Stroke) or TI=(Cerebral Strokes) or TI=(Stroke, Cerebral) or TI=(Strokes, Cerebral) or TI=(Stroke, Acute) or TI=(Acute Stroke) or TI=(Acute Strokes) or TI=(Strokes, Acute) or TI=(Cerebrovascular Accident, Acute) or TI=(Acute Cerebrovascular Accident) or TI=(Acute Cerebrovascular Accidents) or TI=(Cerebrovascular Accidents, Acute)
#2	(TI=(COVID-19)) or TI=(COVID 19) or TI=(COVID-19 Virus Disease) or TI=(COVID 19 Virus Disease) or TI=(COVID-19 Virus Diseases) or TI=(Disease, COVID-19 Virus) or TI=(Virus Disease, COVID-19) or TI=(COVID-19 Virus Infection) or TI=(COVID 19 Virus Infection) or TI=(COVID-19 Virus Infections) or TI=(Infection, COVID-19 Virus) or TI=(Virus Infection, COVID-19) or TI=(2019-nCoV Infection) or TI=(2019 nCoV Infection) or TI=(2019-nCoV Infections) or TI=(Infection, 2019-nCoV) or TI=(Coronavirus Disease-19) or TI=(Coronavirus Disease 19) or TI=(2019 Novel Coronavirus Disease) or TI=(2019 Novel Coronavirus Infection) or TI=(2019-nCoV Disease) or TI=(2019 nCoV Disease) or TI=(2019-nCoV Diseases) or TI=(Disease, 2019-nCoV) or TI=(COVID19) or TI=(Coronavirus Disease 2019) or TI=(Disease 2019, Coronavirus) or TI=(SARS Coronavirus 2 Infection) or TI=(SARS-CoV-2 Infection) or TI=(Infection, SARS-CoV-2) or TI=(SARS CoV 2 Infection) or TI=(SARS-CoV-2 Infections) or TI=(COVID-19 Pandemic) or TI=(COVID 19 Pandemic) or TI=(Pandemic, COVID-19)
#1	TS=(cerebral hemorrhage* or Hemorrhage, Cerebrum* or Cerebrum Hemorrhage* or Cerebrum Hemorrhages* or Hemorrhages, Cerebrum* or Cerebral Parenchymal Hemorrhage* or Cerebral Parenchymal Hemorrhages* or Hemorrhage, Cerebral Parenchymal* or Hemorrhages, Cerebral Parenchymal* or Parenchymal Hemorrhage, Cerebral* or Parenchymal Hemorrhages, Cerebral* or Intracerebral Hemorrhage* or Hemorrhage, Intracerebral* or Hemorrhages, Intracerebral* or Intracerebral Hemorrhages* or Hemorrhage, Cerebral* or Cerebral Hemorrhages* or Hemorrhages, Cerebral* or Brain Hemorrhage, Cerebral* or Brain Hemorrhages, Cerebral* or Cerebral Brain Hemorrhage* or Cerebral Brain Hemorrhages* or Hemorrhage, Cerebral Brain* or Hemorrhages, Cerebral Brain*)
Elsevier search strategy	
#1	(("cerebral hemorrhage" OR"Stroke" OR"subarachnoid hemorrhage" OR"hemorrhagic stroke ")) AND "COVID 19"
Medline search strategy	
#7	((ab(cerebral hemorrhage) OR ab(Hemorrhage, Cerebrum) OR ab(Cerebrum Hemorrhage) OR ab(Cerebrum Hemorrhages) OR ab(Hemorrhages, Cerebrum) OR ab(Cerebral Parenchymal Hemorrhage) OR ab(Cerebral Parenchymal Hemorrhages) OR ab(Hemorrhage, Cerebral Parenchymal) OR ab(Hemorrhages, Cerebral Parenchymal) OR ab(Parenchymal Hemorrhage, Cerebral) OR ab(Parenchymal Hemorrhages, Cerebral) OR ab(Intracerebral Hemorrhage) OR ab(Hemorrhage, Intracerebral) OR ab(Hemorrhages, Intracerebral) OR ab(Intracerebral Hemorrhages) OR ab(Hemorrhage, Cerebral) OR ab(Cerebral Hemorrhages) OR ab(Hemorrhages, Cerebral) OR ab(Brain Hemorrhage, Cerebral) OR ab(Brain Hemorrhages, Cerebral) OR ab(Cerebral Brain Hemorrhage) OR ab(Cerebral Brain Hemorrhages) OR ab(Hemorrhage, Cerebral Brain) OR ab(Hemorrhages, Cerebral Brain)) OR (ab(Stroke) OR ab(Strokes) OR ab(Cerebrovascular Accident) OR ab(Cerebrovascular Accidents) OR ab(CVA (Cerebrovascular Accident)) OR ab(CVAs (Cerebrovascular Accident)) OR ab(Cerebrovascular Apoplexy) OR ab(Apoplexy, Cerebrovascular) OR ab(Vascular Accident, Brain) OR ab(Brain Vascular Accident) OR ab(Brain Vascular Accidents) OR ab(Vascular Accidents, Brain) OR ab(Cerebrovascular Stroke) OR ab(Cerebrovascular Strokes) OR ab(Stroke, Cerebrovascular) OR ab(Strokes, Cerebrovascular) OR ab(Apoplexy) OR ab(Cerebral Stroke) OR ab(Cerebral Strokes) OR ab(Stroke, Cerebral) OR ab(Strokes, Cerebral) OR ab(Stroke, Acute) OR ab(Acute Stroke) OR ab(Acute Strokes) OR ab(Strokes, Acute) OR ab(Cerebrovascular Accident, Acute) OR ab(Acute Cerebrovascular Accident) OR ab(Acute Cerebrovascular Accidents) OR ab(Cerebrovascular Accidents, Acute)) OR (ab(subarachnoid hemorrhage) OR ab(SAH) OR ab(SAHs) OR ab(Hemorrhage, Subarachnoid) OR ab(Hemorrhages, Subarachnoid) OR ab(Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhage, Aneurysmal) OR ab(Aneurysmal Subarachnoid Hemorrhage) OR ab(Aneurysmal Subarachnoid Hemorrhages) OR ab(Hemorrhage, Aneurysmal Subarachnoid) OR ab(Hemorrhages, Aneurysmal Subarachnoid) OR ab(Subarachnoid Hemorrhages, Aneurysmal) OR ab(Subarachnoid Hemorrhage, Spontaneous) OR ab(Hemorrhage, Spontaneous Subarachnoid) OR ab(Hemorrhages, Spontaneous Subarachnoid) OR ab(Spontaneous Subarachnoid Hemorrhage) OR ab(Spontaneous Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhages, Spontaneous) OR ab(Perinatal Subarachnoid Hemorrhage) OR ab(Hemorrhage, Perinatal Subarachnoid) OR ab(Hemorrhages, Perinatal Subarachnoid) OR ab(Perinatal Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhage, Perinatal) OR ab(Subarachnoid Hemorrhages, Perinatal) OR ab(Subarachnoid Hemorrhage, Intracranial) OR ab(Hemorrhage, Intracranial Subarachnoid) OR ab(Hemorrhages, Intracranial Subarachnoid) OR ab(Intracranial Subarachnoid Hemorrhage) OR ab(Intracranial Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhages, Intracranial)) OR (ab(hemorrhagic stroke) OR ab(Hemorrhagic Strokes) OR ab(Stroke, Hemorrhagic) OR ab(Subarachnoid Hemorrhagic Stroke) OR ab(Hemorrhagic Stroke, Subarachnoid) OR ab(Stroke, Subarachnoid Hemorrhagic) OR ab(Subarachnoid Hemorrhagic Strokes) OR ab(Intracerebral Hemorrhagic Stroke) OR ab(Hemorrhagic Stroke, Intracerebral) OR ab(Intracerebral Hemorrhagic Strokes) OR ab(Stroke, Intracerebral Hemorrhagic) OR ab(Intracerebral Hemorrhage Stroke) OR ab(Hemorrhage Stroke, Intracerebral) OR ab(Intracerebral Hemorrhage Strokes) OR ab(Stroke, Intracerebral Hemorrhage))) AND (ab(COVID-19) OR ab(COVID 19) OR ab(COVID-19 Virus Disease) OR ab(COVID 19 Virus Disease) OR ab(COVID-19 Virus Diseases) OR ab(Disease, COVID-19 Virus) OR ab(Virus Disease, COVID-19) OR ab(COVID-19 Virus Infection) OR ab(COVID 19 Virus Infection) OR ab(COVID-19 Virus Infections) OR ab(Infection, COVID-19 Virus) OR ab(Virus Infection, COVID-19) OR ab(2019-nCoV Infection) OR ab(2019 nCoV Infection) OR ab(2019-nCoV Infections) OR ab(Infection, 2019-nCoV) OR ab(Coronavirus Disease-19) OR ab(Coronavirus Disease 19) OR ab(2019 Novel Coronavirus Disease) OR ab(2019 Novel Coronavirus Infection) OR ab(2019-nCoV Disease) OR ab(2019 nCoV Disease) OR ab(2019-nCoV Diseases) OR ab(Disease, 2019-nCoV) OR ab(COVID19) OR ab(Coronavirus Disease 2019) OR ab(Disease 2019, Coronavirus) OR ab(SARS Coronavirus 2 Infection) OR ab(SARS-CoV-2 Infection) OR ab(Infection, SARS-CoV-2) OR ab(SARS CoV 2 Infection) OR ab(SARS-CoV-2 Infections) OR ab(COVID-19 Pandemic) OR ab(COVID 19 Pandemic) OR ab(COVID-19 Pandemics))
#6	(ab(cerebral hemorrhage) OR ab(Hemorrhage, Cerebrum) OR ab(Cerebrum Hemorrhage) OR ab(Cerebrum Hemorrhages) OR ab(Hemorrhages, Cerebrum) OR ab(Cerebral Parenchymal Hemorrhage) OR ab(Cerebral Parenchymal Hemorrhages) OR ab(Hemorrhage, Cerebral Parenchymal) OR ab(Hemorrhages, Cerebral Parenchymal) OR ab(Parenchymal Hemorrhage, Cerebral) OR ab(Parenchymal Hemorrhages, Cerebral) OR ab(Intracerebral Hemorrhage) OR ab(Hemorrhage, Intracerebral) OR ab(Hemorrhages, Intracerebral) OR ab(Intracerebral Hemorrhages) OR ab(Hemorrhage, Cerebral) OR ab(Cerebral Hemorrhages) OR ab(Hemorrhages, Cerebral) OR ab(Brain Hemorrhage, Cerebral) OR ab(Brain Hemorrhages, Cerebral) OR ab(Cerebral Brain Hemorrhage) OR ab(Cerebral Brain Hemorrhages) OR ab(Hemorrhage, Cerebral Brain) OR ab(Hemorrhages, Cerebral Brain)) OR (ab(Stroke) OR ab(Strokes) OR ab(Cerebrovascular Accident) OR ab(Cerebrovascular Accidents) OR ab(CVA (Cerebrovascular Accident)) OR ab(CVAs (Cerebrovascular Accident)) OR ab(Cerebrovascular Apoplexy) OR ab(Apoplexy, Cerebrovascular) OR ab(Vascular Accident, Brain) OR ab(Brain Vascular Accident) OR ab(Brain Vascular Accidents) OR ab(Vascular Accidents, Brain) OR ab(Cerebrovascular Stroke) OR ab(Cerebrovascular Strokes) OR ab(Stroke, Cerebrovascular) OR ab(Strokes, Cerebrovascular) OR ab(Apoplexy) OR ab(Cerebral Stroke) OR ab(Cerebral Strokes) OR ab(Stroke, Cerebral) OR ab(Strokes, Cerebral) OR ab(Stroke, Acute) OR ab(Acute Stroke) OR ab(Acute Strokes) OR ab(Strokes, Acute) OR ab(Cerebrovascular Accident, Acute) OR ab(Acute Cerebrovascular Accident) OR ab(Acute Cerebrovascular Accidents) OR ab(Cerebrovascular Accidents, Acute)) OR (ab(subarachnoid hemorrhage) OR ab(SAH) OR ab(SAHs) OR ab(Hemorrhage, Subarachnoid) OR ab(Hemorrhages, Subarachnoid) OR ab(Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhage, Aneurysmal) OR ab(Aneurysmal Subarachnoid Hemorrhage) OR ab(Aneurysmal Subarachnoid Hemorrhages) OR ab(Hemorrhage, Aneurysmal Subarachnoid) OR ab(Hemorrhages, Aneurysmal Subarachnoid) OR ab(Subarachnoid Hemorrhages, Aneurysmal) OR ab(Subarachnoid Hemorrhage, Spontaneous) OR ab(Hemorrhage, Spontaneous Subarachnoid) OR ab(Hemorrhages, Spontaneous Subarachnoid) OR ab(Spontaneous Subarachnoid Hemorrhage) OR ab(Spontaneous Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhages, Spontaneous) OR ab(Perinatal Subarachnoid Hemorrhage) OR ab(Hemorrhage, Perinatal Subarachnoid) OR ab(Hemorrhages, Perinatal Subarachnoid) OR ab(Perinatal Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhage, Perinatal) OR ab(Subarachnoid Hemorrhages, Perinatal) OR ab(Subarachnoid Hemorrhage, Intracranial) OR ab(Hemorrhage, Intracranial Subarachnoid) OR ab(Hemorrhages, Intracranial Subarachnoid) OR ab(Intracranial Subarachnoid Hemorrhage) OR ab(Intracranial Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhages, Intracranial)) OR (ab(hemorrhagic stroke) OR ab(Hemorrhagic Strokes) OR ab(Stroke, Hemorrhagic) OR ab(Subarachnoid Hemorrhagic Stroke) OR ab(Hemorrhagic Stroke, Subarachnoid) OR ab(Stroke, Subarachnoid Hemorrhagic) OR ab(Subarachnoid Hemorrhagic Strokes) OR ab(Intracerebral Hemorrhagic Stroke) OR ab(Hemorrhagic Stroke, Intracerebral) OR ab(Intracerebral Hemorrhagic Strokes) OR ab(Stroke, Intracerebral Hemorrhagic) OR ab(Intracerebral Hemorrhage Stroke) OR ab(Hemorrhage Stroke, Intracerebral) OR ab(Intracerebral Hemorrhage Strokes) OR ab(Stroke, Intracerebral Hemorrhage))
#5	ab(hemorrhagic stroke) OR ab(Hemorrhagic Strokes) OR ab(Stroke, Hemorrhagic) OR ab(Subarachnoid Hemorrhagic Stroke) OR ab(Hemorrhagic Stroke, Subarachnoid) OR ab(Stroke, Subarachnoid Hemorrhagic) OR ab(Subarachnoid Hemorrhagic Strokes) OR ab(Intracerebral Hemorrhagic Stroke) OR ab(Hemorrhagic Stroke, Intracerebral) OR ab(Intracerebral Hemorrhagic Strokes) OR ab(Stroke, Intracerebral Hemorrhagic) OR ab(Intracerebral Hemorrhage Stroke) OR ab(Hemorrhage Stroke, Intracerebral) OR ab(Intracerebral Hemorrhage Strokes) OR ab(Stroke, Intracerebral Hemorrhage)

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#4	ab(subarachnoid hemorrhage) OR ab(SAH) OR ab(SAHs) OR ab(Hemorrhage, Subarachnoid) OR ab(Hemorrhages, Subarachnoid) OR ab(Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhage, Aneurysmal) OR ab(Aneurysmal Subarachnoid Hemorrhage) OR ab(Aneurysmal Subarachnoid Hemorrhages) OR ab(Hemorrhage, Aneurysmal Subarachnoid) OR ab(Hemorrhages, Aneurysmal Subarachnoid) OR ab(Subarachnoid Hemorrhages, Aneurysmal) OR ab(Subarachnoid Hemorrhage, Spontaneous) OR ab(Hemorrhage, Spontaneous Subarachnoid) OR ab(Hemorrhages, Spontaneous Subarachnoid) OR ab(Spontaneous Subarachnoid Hemorrhage) OR ab(Spontaneous Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhages, Spontaneous) OR ab(Perinatal Subarachnoid Hemorrhage) OR ab(Hemorrhage, Perinatal Subarachnoid) OR ab(Hemorrhages, Perinatal Subarachnoid) OR ab(Perinatal Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhage, Perinatal) OR ab(Subarachnoid Hemorrhages, Perinatal) OR ab(Subarachnoid Hemorrhage, Intracranial) OR ab(Hemorrhage, Intracranial Subarachnoid) OR ab(Hemorrhages, Intracranial Subarachnoid) OR ab(Intracranial Subarachnoid Hemorrhage) OR ab(Intracranial Subarachnoid Hemorrhages) OR ab(Subarachnoid Hemorrhages, Intracranial)
#3	ab(Stroke) OR ab(Strokes) OR ab(Cerebrovascular Accident) OR ab(Cerebrovascular Accidents) OR ab(CVA (Cerebrovascular Accident)) OR ab(CVAs (Cerebrovascular Accident)) OR ab(Cerebrovascular Apoplexy) OR ab(Apoplexy, Cerebrovascular) OR ab(Vascular Accident, Brain) OR ab(Brain Vascular Accident) OR ab(Brain Vascular Accidents) OR ab(Vascular Accidents, Brain) OR ab(Cerebrovascular Stroke) OR ab(Cerebrovascular Strokes) OR ab(Stroke, Cerebrovascular) OR ab(Strokes, Cerebrovascular) OR ab(Apoplexy) OR ab(Cerebral Stroke) OR ab(Cerebral Strokes) OR ab(Stroke, Cerebral) OR ab(Strokes, Cerebral) OR ab(Stroke, Acute) OR ab(Acute Stroke) OR ab(Acute Strokes) OR ab(Strokes, Acute) OR ab(Cerebrovascular Accident, Acute) OR ab(Acute Cerebrovascular Accident) OR ab(Acute Cerebrovascular Accidents) OR ab(Cerebrovascular Accidents, Acute)
#2	ab(COVID-19) OR ab(COVID 19) OR ab(COVID-19 Virus Disease) OR ab(COVID 19 Virus Disease) OR ab(COVID-19 Virus Diseases) OR ab(Disease, COVID-19 Virus) OR ab(Virus Disease, COVID-19) OR ab(COVID-19 Virus Infection) OR ab(COVID 19 Virus Infection) OR ab(COVID-19 Virus Infections) OR ab(Infection, COVID-19 Virus) OR ab(Virus Infection, COVID-19) OR ab(2019-nCoV Infection) OR ab(2019 nCoV Infection) OR ab(2019-nCoV Infections) OR ab(Infection, 2019-nCoV) OR ab(Coronavirus Disease-19) OR ab(Coronavirus Disease 19) OR ab(2019 Novel Coronavirus Disease) OR ab(2019 Novel Coronavirus Infection) OR ab(2019-nCoV Disease) OR ab(2019 nCoV Disease) OR ab(2019-nCoV Diseases) OR ab(Disease, 2019-nCoV) OR ab(COVID19) OR ab(Coronavirus Disease 2019) OR ab(Disease 2019, Coronavirus) OR ab(SARS Coronavirus 2 Infection) OR ab(SARS-CoV-2 Infection) OR ab(Infection, SARS-CoV-2) OR ab(SARS CoV 2 Infection) OR ab(SARS-CoV-2 Infections) OR ab(COVID-19 Pandemic) OR ab(COVID 19 Pandemic) OR ab(COVID-19 Pandemics)
#1	ab(cerebral hemorrhage) OR ab(Hemorrhage, Cerebrum) OR ab(Cerebrum Hemorrhage) OR ab(Cerebrum Hemorrhages) OR ab(Hemorrhages, Cerebrum) OR ab(Cerebral Parenchymal Hemorrhage) OR ab(Cerebral Parenchymal Hemorrhages) OR ab(Hemorrhage, Cerebral Parenchymal) OR ab(Hemorrhages, Cerebral Parenchymal) OR ab(Parenchymal Hemorrhage, Cerebral) OR ab(Parenchymal Hemorrhages, Cerebral) OR ab(Intracerebral Hemorrhage) OR ab(Hemorrhage, Intracerebral) OR ab(Hemorrhages, Intracerebral) OR ab(Intracerebral Hemorrhages) OR ab(Hemorrhage, Cerebral) OR ab(Cerebral Hemorrhages) OR ab(Hemorrhages, Cerebral) OR ab(Brain Hemorrhage, Cerebral) OR ab(Brain Hemorrhages, Cerebral) OR ab(Cerebral Brain Hemorrhage) OR ab(Cerebral Brain Hemorrhages) OR ab(Hemorrhage, Cerebral Brain) OR ab(Hemorrhages, Cerebral Brain)
Cochrane Library search strategy	
#7	#6 AND #2
#6	#1 OR #3 OR #4 OR #5
#5	(hemorrhagic stroke):ti,ab,kw OR (Hemorrhagic Strokes):ti,ab,kw OR (Stroke, Hemorrhagic):ti,ab,kw OR (Subarachnoid Hemorrhagic Stroke):ti,ab,kw OR (Hemorrhagic Stroke, Subarachnoid):ti,ab,kw OR (Stroke, Subarachnoid Hemorrhagic):ti,ab,kw OR (Subarachnoid Hemorrhagic Strokes):ti,ab,kw OR (Intracerebral Hemorrhagic Stroke):ti,ab,kw OR (Hemorrhagic Stroke, Intracerebral):ti,ab,kw OR (Intracerebral Hemorrhagic Strokes):ti,ab,kw OR (Stroke, Intracerebral Hemorrhagic):ti,ab,kw OR (Intracerebral Hemorrhage Stroke):ti,ab,kw OR (Hemorrhage Stroke, Intracerebral):ti,ab,kw OR (Intracerebral Hemorrhage Strokes):ti,ab,kw OR (Stroke, Intracerebral Hemorrhage):ti,ab,kw
#4	(subarachnoid hemorrhage):ti,ab,kw OR (SAH):ti,ab,kw OR (SAHs):ti,ab,kw OR (Hemorrhage, Subarachnoid):ti,ab,kw OR (Hemorrhages, Subarachnoid):ti,ab,kw OR (Subarachnoid Hemorrhages):ti,ab,kw OR (Subarachnoid Hemorrhage, Aneurysmal):ti,ab,kw OR (Aneurysmal Subarachnoid Hemorrhage):ti,ab,kw OR (Aneurysmal Subarachnoid Hemorrhages):ti,ab,kw OR (Hemorrhage, Aneurysmal Subarachnoid):ti,ab,kw OR (Hemorrhages, Aneurysmal Subarachnoid):ti,ab,kw OR (Subarachnoid Hemorrhages, Aneurysmal):ti,ab,kw OR (Subarachnoid Hemorrhage, Spontaneous):ti,ab,kw OR (Hemorrhage, Spontaneous Subarachnoid):ti,ab,kw OR (Hemorrhages, Spontaneous Subarachnoid):ti,ab,kw OR (Spontaneous Subarachnoid Hemorrhage):ti,ab,kw OR (Spontaneous Subarachnoid Hemorrhages):ti,ab,kw OR (Subarachnoid Hemorrhages, Spontaneous):ti,ab,kw OR (Perinatal Subarachnoid Hemorrhage):ti,ab,kw OR (Hemorrhage, Perinatal Subarachnoid):ti,ab,kw OR (Hemorrhages, Perinatal Subarachnoid):ti,ab,kw OR (Perinatal Subarachnoid Hemorrhages):ti,ab,kw OR (Subarachnoid Hemorrhage, Perinatal):ti,ab,kw OR (Subarachnoid Hemorrhages, Perinatal):ti,ab,kw OR (Subarachnoid Hemorrhage, Intracranial):ti,ab,kw OR (Hemorrhage, Intracranial Subarachnoid):ti,ab,kw OR (Hemorrhages, Intracranial Subarachnoid):ti,ab,kw OR (Intracranial Subarachnoid Hemorrhage):ti,ab,kw OR (Intracranial Subarachnoid Hemorrhages):ti,ab,kw OR (Subarachnoid Hemorrhages, Intracranial):ti,ab,kw
#3	(Stroke):ti,ab,kw OR (Strokes):ti,ab,kw OR (Cerebrovascular Accident):ti,ab,kw OR (Cerebrovascular Accidents):ti,ab,kw OR (CVA (Cerebrovascular Accident)):ti,ab,kw OR (CVAs (Cerebrovascular Accident)):ti,ab,kw OR (Cerebrovascular Apoplexy):ti,ab,kw OR (Apoplexy, Cerebrovascular):ti,ab,kw OR (Vascular Accident, Brain):ti,ab,kw OR (Brain Vascular Accident):ti,ab,kw OR (Brain Vascular Accidents):ti,ab,kw OR (Vascular Accidents, Brain):ti,ab,kw OR (Cerebrovascular Stroke):ti,ab,kw OR (Cerebrovascular Strokes):ti,ab,kw OR (Stroke, Cerebrovascular):ti,ab,kw OR (Strokes, Cerebrovascular):ti,ab,kw OR (Apoplexy):ti,ab,kw OR (Cerebral Stroke):ti,ab,kw OR (Cerebral Strokes):ti,ab,kw OR (Stroke, Cerebral):ti,ab,kw OR (Strokes, Cerebral):ti,ab,kw OR (Stroke, Acute):ti,ab,kw OR (Acute Stroke):ti,ab,kw OR (Acute Strokes):ti,ab,kw OR (Strokes, Acute):ti,ab,kw OR (Cerebrovascular Accident, Acute):ti,ab,kw OR (Acute Cerebrovascular Accident):ti,ab,kw OR (Acute Cerebrovascular Accidents):ti,ab,kw OR (Cerebrovascular Accidents, Acute):ti,ab,kw
#2	(COVID 19):ti,ab,kw OR (COVID 19):ti,ab,kw OR (COVID 19 Virus Disease):ti,ab,kw OR (COVID 19 Virus Disease):ti,ab,kw OR (COVID 19 Virus Diseases):ti,ab,kw OR (Disease, COVID 19 Virus):ti,ab,kw OR (Virus Disease, COVID 19):ti,ab,kw OR (COVID 19 Virus Infection):ti,ab,kw OR (COVID 19 Virus Infection):ti,ab,kw OR (COVID 19 Virus Infections):ti,ab,kw OR (Infection, COVID 19 Virus):ti,ab,kw OR (Virus Infection, COVID 19):ti,ab,kw OR (2019 nCoV Infection):ti,ab,kw OR (2019 nCoV Infection):ti,ab,kw OR (2019 nCoV Infections):ti,ab,kw OR (Infection, 2019 nCoV):ti,ab,kw OR (Coronavirus Disease 19):ti,ab,kw OR (Coronavirus Disease 19):ti,ab,kw OR (2019 Novel Coronavirus Disease):ti,ab,kw OR (2019 Novel Coronavirus Infection):ti,ab,kw OR (2019 nCoV Disease):ti,ab,kw OR (2019 nCoV Disease):ti,ab,kw OR (2019 nCoV Diseases):ti,ab,kw OR (Disease, 2019 nCoV):ti,ab,kw OR (COVID19):ti,ab,kw OR (Coronavirus Disease 2019):ti,ab,kw OR (Disease 2019, Coronavirus):ti,ab,kw OR (SARS Coronavirus 2 Infection):ti,ab,kw OR (SARS CoV 2 Infection):ti,ab,kw OR (Infection, SARS CoV 2):ti,ab,kw OR (SARS CoV 2 Infection):ti,ab,kw OR (SARS CoV 2 Infections):ti,ab,kw OR (COVID 19 Pandemic):ti,ab,kw OR (COVID 19 Pandemic):ti,ab,kw OR (COVID 19 Pandemics):ti,ab,kw OR (Pandemic, COVID 19):ti,ab,kw
#1	(cerebral hemorrhage):ti,ab,kw OR (Hemorrhage, Cerebrum):ti,ab,kw OR (Cerebrum Hemorrhage):ti,ab,kw OR (Cerebrum Hemorrhages):ti,ab,kw OR (Hemorrhages, Cerebrum):ti,ab,kw OR (Cerebral Parenchymal Hemorrhage):ti,ab,kw OR (Cerebral Parenchymal Hemorrhages):ti,ab,kw OR (Hemorrhage, Cerebral Parenchymal):ti,ab,kw OR (Hemorrhages, Cerebral Parenchymal):ti,ab,kw OR (Parenchymal Hemorrhage, Cerebral):ti,ab,kw OR (Parenchymal Hemorrhages, Cerebral):ti,ab,kw OR (Intracerebral Hemorrhage):ti,ab,kw OR (Hemorrhage, Intracerebral):ti,ab,kw OR (Hemorrhages, Intracerebral):ti,ab,kw OR (Intracerebral Hemorrhages):ti,ab,kw OR (Hemorrhage, Cerebral):ti,ab,kw OR (Cerebral Hemorrhages):ti,ab,kw OR (Hemorrhages, Cerebral):ti,ab,kw OR (Brain Hemorrhage, Cerebral):ti,ab,kw OR (Brain Hemorrhages, Cerebral):ti,ab,kw OR (Cerebral Brain Hemorrhage):ti,ab,kw OR (Cerebral Brain Hemorrhages):ti,ab,kw OR (Hemorrhage, Cerebral Brain):ti,ab,kw OR (Hemorrhages, Cerebral Brain):ti,ab,kw
Google scholar search strategy	
#1	allintitle: COVID19 "subarachnoid hemorrhage" OR "cerebral hemorrhage" OR "stroke"

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7-8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8-9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11-13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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